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BACHELOR THESIS IN COMPUTER SCIENCE

Robust Spanning Trees

under Discrete and Interval Uncertainty
Min-Max and Min-Max Regret Models

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Abstract

The minimum spanning tree (MST) connects all nodes of a network at minimum cost, but those costs are rarely known when the network must be designed. This thesis studies robust spanning trees, fixed in advance and judged against every cost the data allows. It pairs two objectives, min-max and min-max regret, with three uncertainty models: discrete scenarios, per-edge intervals, and a budget capping the total deviation from nominal costs. Each of the six combinations is classified by computational complexity and by the approximation guarantees available.

Under interval costs the upper bounds are worst for every tree at once, so min-max reduces to a single deterministic MST computation. Min-max regret is strongly NP-hard there, and approximable within a factor of 2. Under a budget min-max stays polynomial, reducing to at most $|E| + 1$ deterministic MST computations, while min-max regret inherits that hardness, which settles the sixth combination. Under discrete scenarios the two objectives behave alike. Both are weakly NP-hard already at two scenarios, and admit pseudo-polynomial algorithms and an FPTAS for constant K . Once K is part of the input both turn strongly NP-hard, and neither is approximable within $2 - \varepsilon$ unless $P = NP$. They part only over the guarantees available there, a logarithmic factor for min-max against K for regret.

We prove fourteen results in full: the five MST foundations, the interval extremal characterisations, the budgeted reformulation, and both two-scenario reductions. Others, among them the interval hardness and the midpoint guarantee, are developed from their sources in the text; the rest is stated on the authority of the literature. A classification table sets the results side by side, and a fixed four-vertex micro-graph runs through the thesis, closing in a gallery where the two objectives select different trees. A single criterion explains the picture: complexity is set by the cost of making the adversary's worst case independent of the chosen tree.

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Chapter 1

Introduction

1.1 Motivation

Consider the task of designing a fibre-optic telecommunications network connecting major cities across a region. The network must link all locations while minimising the total cost of laying cable. That cost is not known when the design is fixed: it depends on terrain, since cable through mountains costs far more than cable across plains, on regulatory approvals that vary by jurisdiction, and on material prices that move with global supply chains. Uncertainty of this kind is pervasive in infrastructure planning, and as Kouvelis and Yu [1] observe, forecast parameter values rarely match realised costs in practice. A design that looks optimal under today’s estimates can prove expensive under the costs that actually arrive, which raises the question this thesis takes up: how should a network be chosen when its costs become known only after the choice is made?

When the costs are known precisely, the *minimum spanning tree* (MST) answers the question. It connects all nodes at minimum total cost and is computed efficiently by well-established algorithms such as Kruskal’s greedy edge selection or Prim’s incremental tree growing, both running in polynomial time [2, Chapter 6]. When they are not, the MST of any one estimate carries no guarantee for the others: it may cost much more than expected once the true costs arrive, or lose out to a tree that a different estimate would have favoured. The deterministic problem is therefore well understood and, on its own, not enough: what is wanted is a tree that is *robust* to cost uncertainty.

Under uncertainty, two natural design objectives emerge. The first is the *min-max* objective, which minimises the worst-case absolute cost over all cost realisations the model admits. It is the conservative choice, selecting a spanning tree whose cost stays controlled even in the most unfavourable circumstances.

The second is *min-max regret*, which judges a tree not in absolute terms but against what could have been achieved with hindsight. The regret of a tree in a given realisation is the difference between its cost and the cost of the best tree for that realisation. Minimising the worst-case regret guards against a different failure: not a high absolute cost, but a choice that a better-informed planner would not have made.

This thesis models the uncertainty in three ways. Under *discrete scenario uncertainty* a planner enumerates a finite list of plausible cost vectors, each a possible

future state; in the telecommunications example, these might be combinations of regulatory outcomes and material price levels. Under *interval uncertainty* each edge cost is known only to lie between a lower and an upper estimate, which suits costs that vary continuously within fixed limits, such as commodity prices or exchange rates. Under *budgeted uncertainty* every edge again has an interval, but a single parameter caps the total deviation across the graph, letting the modeller choose how conservative the analysis should be rather than commit to the full ranges at once. That parameter interpolates: at one end the problem is the deterministic one on the nominal costs, at the other it is the interval model in full.

Two objectives and three uncertainty models give six combinations, and this thesis classifies every one of them. The central question is this: how do computational complexity and approximability vary across the six? [Section 1.2](#) makes the question precise and fixes the scope within which it is answered.

1.2 Research Questions and Scope

That question divides into three.

First, what is the computational complexity of the six combinations? Does each admit a polynomial-time algorithm, or is it intractable? Where hardness arises, does it come from the number of scenarios being unbounded, or is two scenarios already enough? Answering this requires precise classifications: polynomial-time solvability, weak NP-hardness, which admits pseudo-polynomial algorithms, or strong NP-hardness, which remains hard even when every numeric value in the instance is bounded by a polynomial in the input size.

Second, where does the worst case lie inside an interval uncertainty set, and does locating it make the problem easy? Each edge cost ranges over an interval, so the scenarios are infinitely many; if the worst of them can be named in advance, evaluating a single tree needs no search over that infinity. Whether cheap evaluation also makes the minimisation over trees cheap is a further question, and the two objectives answer it differently. Extremal characterisations settle the first half for both, and this thesis proves them.

Third, for problems that are computationally hard, what approximation guarantees can be achieved? Can a polynomial-time algorithm come within a constant factor of the optimum, or within any accuracy demanded of it, as a fully polynomial-time approximation scheme (FPTAS) does? Conversely, are there hardness-of-approximation results that rule out such guarantees? A limit on what any algorithm can do is as much part of the classification as an algorithm is.

The scope of this thesis is deliberately focused. We restrict attention to spanning tree problems in undirected, connected graphs with static, single-stage decisions: a tree is selected once, before any uncertainty is resolved, and remains fixed. The three uncertainty models just described are defined formally in [Section 2.4](#), and the budgeted model is analysed in [Section 3.4](#). No objectives beyond min-max and min-max regret are considered.

This thesis does not cover polyhedral uncertainty sets, ellipsoidal uncertainty, or distributional robustness, where costs follow known probability distributions. It also excludes two-stage and recoverable models, in which decisions can be partly adjusted once realisations are observed. [Section 5.3](#) returns to the last two of these, and to

a generalisation of the budgeted model that replaces its single cap by several; the others lie outside this work entirely. The aim is not encyclopaedic coverage but a self-contained treatment of the models that are in scope.

1.3 Contributions

This thesis delivers six principal contributions. The first five gather established results of the field; the contribution there is the development, with fourteen results proved in full and the remainder developed from their sources or cited at the point of use. The sixth is the thesis’s own synthesis: the classification that draws them together, and the criterion that explains it.

First, we provide self-contained minimum spanning tree foundations ([Chapter 2](#)). We prove five fundamental results from first principles: the fundamental cycle and cut lemmas via exchange arguments, and three MST optimality criteria (cycle property, cut property, and their equivalence). [Chapter 2](#) also includes a complexity primer defining polynomial time, NP-hardness (weak and strong), and approximation concepts (constant-factor algorithms, FPTAS, PTAS) for classifying the robust variants.

Second, we establish extremal characterisations for interval uncertainty ([Chapters 3 and 4](#)). For min-max, [Lemma 3.1](#) shows that the vector of upper bounds is a worst case for every tree at once, reducing the problem to one deterministic MST computation. For regret, [Lemma 4.1](#) names a worst case for each individual tree, charging every edge of that tree its upper bound and every other edge its lower bound. That worst case therefore moves with the tree, and the minimisation over trees stays hard. Both lemmas carry complete proofs, as do the two corollaries that draw their algorithmic consequences.

Third, we prove that min-max stays polynomial under budgeted uncertainty ([Section 3.4](#)). The worst case again depends on the tree here, since the budget is spent on the edges of the chosen tree, and [Theorem 3.6](#) removes that dependence cheaply: fixing a single deviation level restores an edge-by-edge cost for every tree, and at most $|E| + 1$ levels need to be tried. The result is due to Bertsimas and Sim [3]; the two supporting lemmas and the theorem are proved here in full.

Fourth, we give two representative hardness proofs ([Chapters 3 and 4](#)). [Theorem 3.1](#) establishes weak NP-hardness of min-max spanning trees with two discrete scenarios by reduction from the partition problem, on a grid graph whose scenario costs encode target subset sums; the proof carries the construction, the correctness argument and the encoding analysis. [Theorem 4.3](#) reuses that construction unaltered and adjusts only the analysis. Under discrete scenarios the two objectives carry the same complexity at every scenario count, though once K is part of the input they no longer share an approximation guarantee.

Fifth, we present the 2-approximation for interval regret spanning trees ([Chapter 4](#)). [Theorem 4.2](#) states that a minimum spanning tree under the midpoint costs has maximum regret at most twice that of an optimal tree; the algorithm is due to Kasperski and Zieliński [4], and its argument is developed in the text rather than proved. No better guarantee is known for any combinatorial problem of this kind. Whether any algorithm can improve on the factor 2, for a min-max regret problem

with interval costs whose deterministic version is solvable in polynomial time, remains an open question in the literature [5, Open Problem 2].

Sixth, we draw the results together (Chapter 5). Table 5.1 records each combination with its complexity class and the guarantees available for it, and the one combination Chapter 4 leaves aside, min-max regret under a budget, is settled there, its strong hardness inherited from the interval case. Across the table min-max and regret separate on complexity under intervals and under a budget alike. Section 5.1 accounts for that with a single criterion: what decides complexity is the price of making the adversary’s worst case independent of the chosen tree. That price is nothing under intervals and one enumerated level under a budget, and under discrete scenarios it cannot be paid at all unless $P = NP$.

A fixed four-vertex micro-graph runs through the thesis, carrying interval costs, the discrete scenarios read off them, and budgeted parameters drawn from the same data. Section 5.2 solves it under every combination the thesis develops. The two objectives disagree on this instance: min-max returns one tree and regret another in every panel, and reading every interval at its midpoint and solving the deterministic problem returns the regret answer, not the min-max one. Figure 5.2 places all eight spanning trees by worst-case cost and maximum regret, and shows that only those two are ever selected.

1.4 Thesis Structure

The remainder of this thesis is organised as follows.

Chapter 2 establishes mathematical foundations, proving minimum spanning tree optimality criteria from first principles and defining the robust optimisation framework (uncertainty models, objectives, complexity terminology).

Chapter 3 analyses the min-max objective across all three uncertainty models, characterising worst-case costs under intervals, establishing computational complexity for discrete scenarios across different values of the scenario count parameter, and proving polynomial solvability under a budget.

Chapter 4 examines the min-max regret objective, which measures relative performance rather than absolute cost. Interval regret is shown to be harder than interval min-max despite sharing extremal properties, while discrete regret mirrors the complexity hierarchy of discrete min-max.

Chapter 5 sets the two objectives side by side: it consolidates the results of Chapters 3 and 4 into a classification table and solves the running micro-graph in an example gallery. It then answers the question posed in Section 1.1 and turns, in Section 5.3, to what lies beyond the scope fixed above and to two open questions the literature leaves standing.

Chapter A collects the notation of the thesis in a single table, grouped by category, with the section introducing each symbol.

Chapter 2

Foundations

This chapter develops the foundations needed for analysing robust spanning tree problems. After fixing graph-theoretic notation and introducing the running micro-graph (Section 2.1), we prove the MST optimality criteria from first principles (Section 2.2) and remark on the classical greedy algorithms they justify (Section 2.3). Section 2.4 then formalises the three uncertainty models, discrete scenarios, interval uncertainty, and budgeted uncertainty, together with the min-max and min-max regret objectives. Section 2.5 reviews the complexity and approximation terminology used in Chapters 3 and 4.

2.1 Graphs, Trees, and Notation

We work exclusively with undirected graphs throughout this thesis. A *graph* is a pair $G = (V, E)$, where V is a finite non-empty set of *vertices* (or *nodes*) and $E \subseteq \{\{u, v\} : u, v \in V, u \neq v\}$ is a set of *edges*. Each edge connects exactly two distinct vertices; we write $\{u, v\} \in E$ or equivalently $uv \in E$ to denote an edge between u and v . A graph is called *simple* if it contains no multiple edges (at most one edge between any pair of vertices) and no loops (edges connecting a vertex to itself). All graphs considered in this thesis are simple. We denote $n = |V|$ and $m = |E|$ for the number of vertices and edges, respectively. Two vertices $u, v \in V$ are *adjacent* (or *neighbours*) if $uv \in E$, and the *degree* of a vertex v , written $\deg(v)$, is the number of edges incident to v .

A *path* in G is a sequence of distinct vertices $v_0, v_1, \dots, v_k$ such that $\{v_{i-1}, v_i\} \in E$ for all $i \in \{1, \dots, k\}$. The *length* of this path is k (the number of edges). A *cycle* is a sequence of distinct vertices $v_0, v_1, \dots, v_k$ with $k \geq 2$ such that $\{v_{i-1}, v_i\} \in E$ for all $i \in \{1, \dots, k\}$ and $\{v_k, v_0\} \in E$.

A graph G is *connected* if for every pair of vertices $u, v \in V$, there exists a path from u to v . A spanning tree can only exist if G is connected, since the tree must reach every vertex. **Throughout this thesis, we assume all graphs are connected.**

For a non-empty proper subset $X \subseteq V$ (that is, $\emptyset \neq X \subsetneq V$), the *cut* induced by X is the set of edges crossing the partition $(X, V \setminus X)$:

$$\delta(X) = \{\{u, v\} \in E : u \in X, v \in V \setminus X\}.$$

Intuitively, $\delta(X)$ comprises all edges with exactly one endpoint in X . Figure 2.1 shows the case $X = \{1, 2\}$: the three red edges each have exactly one endpoint in X and therefore belong to $\delta(X)$. The edge $\{1, 2\}$ lies wholly inside X and the edge $\{3, 4\}$ wholly outside, so neither of those two crosses the partition.

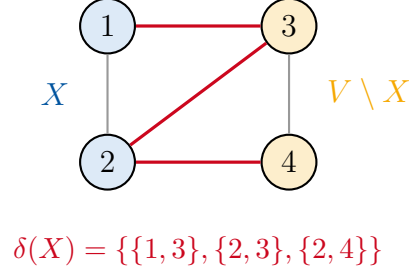


Figure 2.1: Illustration of a cut $\delta(X)$ induced by vertex set $X = \{1, 2\}$. The cut $\delta(X)$ consists of the three red edges connecting the blue-shaded vertices in X to the orange-shaded vertices in $V \setminus X$.

Each edge $e \in E$ has an associated *cost* $c_e \in \mathbb{R}$. We represent the cost structure either as a *cost function* $c: E \rightarrow \mathbb{R}$ or equivalently as a *cost vector* $c \in \mathbb{R}^{|E|}$ indexed by the edges. Costs may be positive, negative, or zero in the deterministic setting; the robust models in Section 2.4 will additionally impose non-negative costs. For any subset $F \subseteq E$, we write $c(F) = \sum_{e \in F} c_e$ for the total cost of the edge set F .

Spanning Trees. A *spanning tree* of G is a connected acyclic subgraph $T \subseteq G$ that includes all vertices of G , that is, $V(T) = V(G)$. We denote by $\mathcal{T}$ the set of all spanning trees of G . Since spanning trees are both connected and acyclic, they satisfy precisely $|E(T)| = n - 1$ edges, where $E(T) \subseteq E$ denotes the edge set of T . These three properties, connectedness, acyclicity, and having exactly $n - 1$ edges, are not independent: for a subgraph spanning all n vertices, any two of them imply the third [2, Theorem 2.1]. We use this fact repeatedly when verifying that a constructed subgraph is a spanning tree. For a spanning tree $T \in \mathcal{T}$, we write $c(T) = c(E(T)) = \sum_{e \in E(T)} c_e$ for the cost of T .

The *minimum spanning tree problem* (MST) seeks a spanning tree of minimum cost:

$$\text{MST}(c) = \min_{T \in \mathcal{T}} c(T). \quad (2.1)$$

Any tree $T^* \in \mathcal{T}$ satisfying $c(T^*) = \text{MST}(c)$ is called an *MST* under cost vector c . The MST is unique if all edge costs are distinct; when edge costs have ties (equal values), multiple spanning trees may achieve the minimum cost $\text{MST}(c)$.

A spanning tree T possesses three fundamental structural properties that we exploit repeatedly:

- (P1) For any two vertices $u, v \in V$, there exists a *unique* path in T connecting u and v .
- (P2) Removing any edge $e \in E(T)$ disconnects T into exactly two connected components.

(P3) Adding any edge $f \in E \setminus E(T)$ to T creates exactly one cycle.

Property **P1** follows from the definition of trees as connected acyclic graphs: connectivity ensures path existence, while acyclicity guarantees uniqueness (a second path would form a cycle). Properties **P2** and **P3** are immediate consequences: removing an edge from a minimally connected subgraph (a tree on n vertices has exactly $n - 1$ edges) must disconnect it, while adding an edge to a maximally acyclic subgraph must create a cycle [2, Theorem 2.1].

Fundamental Structures. We now introduce two graph-theoretic constructs, fundamental cycles and fundamental cuts, that play a central role in characterising MST optimality. Let $T \in \mathcal{T}$ be a spanning tree and let $f \in E \setminus E(T)$ be an edge not in T with endpoints u and v . By property **P3**, adding f to T creates a unique cycle. Since T is a tree, by property **P1**, there exists a unique path P in T from u to v . We call the cycle $C_f = E(P) \cup \{f\}$ (where $E(P)$ denotes the edges of path P) the *fundamental cycle* of f with respect to T .

Similarly, let $e \in E(T)$ be an edge in the tree T . By property **P2**, removing e from T partitions V into exactly two connected components. Let $X_e \subseteq V$ denote one of these components (the choice is arbitrary; either component induces the same cut). The *fundamental cut* of e with respect to T is the cut $\delta(X_e)$ induced by X_e . Crucially, e is the *only* tree edge in $\delta(X_e)$; all other edges in the cut belong to $E \setminus E(T)$. This follows because X_e is a connected component of $T \setminus \{e\}$: any other tree edge crossing the partition would connect X_e to $V \setminus X_e$, contradicting the component structure. These fundamental structures underpin the MST optimality criteria developed in Section 2.2: fundamental cycles correspond to potential edge exchanges for non-tree edges, while fundamental cuts correspond to exchanges for tree edges.

Running Example: The Micro-graph. To anchor the concepts of this thesis, we fix a four-vertex graph that recurs throughout as a running example; it is the graph already drawn in Figure 2.1. Let $G = (V, E)$ with vertex set $V = \{1, 2, 3, 4\}$ and five edges:

$$\begin{aligned} e_1 &= \{1, 2\}, & e_2 &= \{2, 3\}, & e_3 &= \{2, 4\}, \\ e_4 &= \{3, 4\}, & e_5 &= \{1, 3\}. \end{aligned}$$

The structure is depicted in Figure 2.2. Vertex 2 serves as a central hub with edges to vertices 1, 3, and 4, while edges $e_5 = \{1, 3\}$ and $e_4 = \{3, 4\}$ provide alternative connections.

In anticipation of the robust optimisation framework of Section 2.4, we associate each edge e_i with an *interval cost* $[\ell_{e_i}, u_{e_i}]$ representing uncertainty in edge costs:

$$\begin{aligned} e_1 &: [2, 4], & e_2 &: [3, 5], & e_3 &: [1, 7], \\ e_4 &: [4, 6], & e_5 &: [5, 7]. \end{aligned}$$

From these intervals, we derive three *discrete scenarios* that instantiate specific cost realisations: $c^{(1)} = (2, 3, 1, 4, 5)$ (all lower bounds), $c^{(2)} = (4, 5, 7, 6, 7)$ (all upper bounds), and $c^{(3)} = (3, 4, 4, 5, 6)$ (interval midpoints). Graph G admits eight

spanning trees in total. We highlight three representative trees that exhibit different structural properties: $T_1 = \{e_1, e_2, e_3\}$ (star centred at vertex 2), $T_2 = \{e_1, e_2, e_4\}$ (path 1–2–3–4), and $T_3 = \{e_2, e_3, e_5\}$ (path 1–3–2–4). These trees and scenarios provide a consistent test bed for demonstrating how optimal solutions vary across objectives and uncertainty models in subsequent chapters.

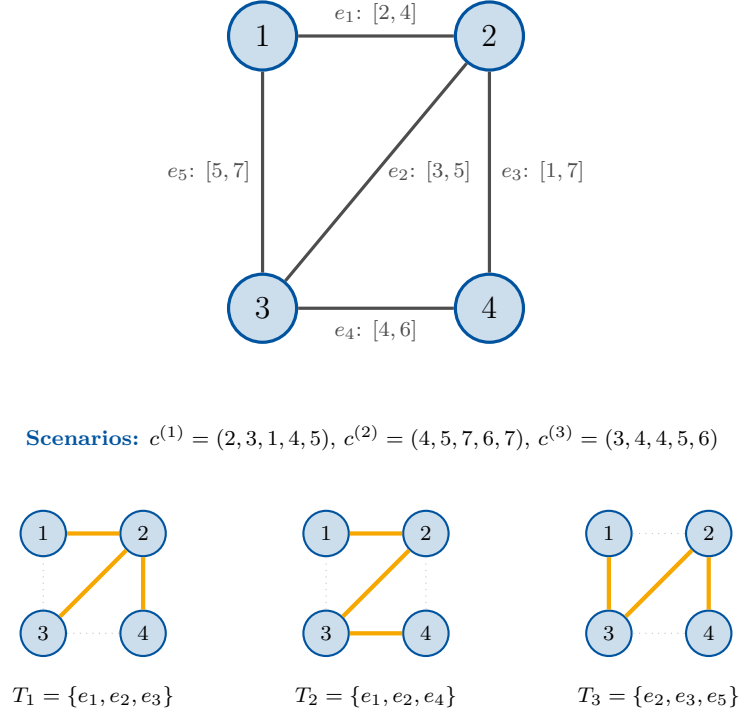


Figure 2.2: The micro-graph G with interval costs $[\ell_e, u_e]$ on each edge (top), used throughout this thesis, together with the three representative spanning trees T_1, T_2, T_3 (bottom, highlighted edges).

2.2 MST Optimality Criteria

A remarkable feature of minimum spanning trees is that optimality can be verified locally: rather than comparing a candidate tree against all exponentially many spanning trees, we can check simple conditions on individual edges. This section develops these local characterisations through five results. We begin with two fundamental exchange lemmas ([Lemmas 2.1](#) and [2.2](#)) that establish necessary conditions for optimality, then prove three equivalent optimality criteria ([Theorems 2.1](#) to [2.3](#)) that provide both necessary and sufficient conditions, thereby fully characterising MST optimality. These characterisations underpin the correctness of classical MST algorithms such as Kruskal’s and Prim’s greedy methods.

Our first lemma relates non-tree edges to the cycles they create.

Lemma 2.1 (Fundamental Cycle Lemma). *Let $G = (V, E)$ be a connected graph with cost function $c: E \rightarrow \mathbb{R}$, and let T be a minimum spanning tree of G under c . For any edge $f \in E \setminus E(T)$, the edge f has maximum cost among all edges in its fundamental cycle C_f . That is, $c_f \geq c_e$ for every edge $e \in C_f$.*

Proof. Suppose for contradiction that f does not have maximum cost in C_f . Then there exists an edge $e \in C_f$ with $c_e > c_f$. Since $f \notin E(T)$ and C_f is created by adding f to T , the edge e must be a tree edge.

We construct an alternative spanning tree by exchanging e for f . Removing e from T disconnects T into two components; since e lies on the path connecting the endpoints of f in T , the edge f bridges these components. Define T' with edge set $E(T') = E(T) \setminus \{e\} \cup \{f\}$.

To verify that T' is a spanning tree, it suffices to check two of the three defining properties. First, $|E(T')| = |E(T)| - 1 + 1 = n - 1$, as required for a tree on n vertices. Second, T' is connected: removing e partitions V into two components, but f crosses this partition (as f completes the cycle C_f containing e), so adding f reconnects the graph. Having $n - 1$ edges and being connected, T' is a spanning tree.

The cost of T' is

$$c(T') = c(T) - c_e + c_f < c(T),$$

where the inequality holds because $c_e > c_f$. This contradicts the optimality of T . Therefore, f must have maximum cost in C_f . $\square$

An analogous statement holds for tree edges, with the fundamental cut $\delta(X_e)$ (introduced in [Section 2.1](#)) playing the role that the fundamental cycle played for non-tree edges.

Lemma 2.2 (Fundamental Cut Lemma). *Let $G = (V, E)$ be a connected graph with cost function $c: E \rightarrow \mathbb{R}$, and let T be a minimum spanning tree of G under c . For any edge $e \in E(T)$, the edge e has minimum cost among all edges in its fundamental cut $\delta(X_e)$. That is, $c_e \leq c_f$ for every edge $f \in \delta(X_e)$.*

Proof. Suppose for contradiction that e does not have minimum cost in $\delta(X_e)$. Then there exists an edge $f \in \delta(X_e)$ with $c_f < c_e$. By construction of the fundamental cut ([Section 2.1](#)), edge e is the unique tree edge in $\delta(X_e)$, so f must be a non-tree edge.

We construct an alternative spanning tree by exchanging e for f . Removing e from T splits V into components X_e and $V \setminus X_e$, which f bridges (since $f \in \delta(X_e)$). Define T' with edge set $E(T') = E(T) \setminus \{e\} \cup \{f\}$.

To verify that T' is a spanning tree, it suffices to check two of the three defining properties. First, $|E(T')| = |E(T)| - 1 + 1 = n - 1$, as required for a tree on n vertices. Second, T' is connected: f has one endpoint in each component (by definition of $\delta(X_e)$), so adding f restores connectivity. Having $n - 1$ edges and being connected, T' is a spanning tree.

The cost of T' is

$$c(T') = c(T) - c_e + c_f < c(T),$$

where the inequality holds because $c_f < c_e$. This contradicts the optimality of T . Therefore, e must have minimum cost in $\delta(X_e)$. $\square$

These fundamental exchange properties establish necessary conditions for a tree to be an MST. The optimality criteria developed next strengthen these to necessary and sufficient characterisations.

Remark 2.1 (Cost Ties). The fundamental lemmas use non-strict inequalities ($c_f \geq c_e$ in [Lemma 2.1](#); $c_e \leq c_f$ in [Lemma 2.2](#)), since multiple edges may share the same cost. When all edge costs are distinct, the MST is unique and the inequalities can be tightened to strict; when ties exist, multiple spanning trees may achieve the minimum cost, and the lemmas apply to each of them.

Theorem 2.1 (Cycle Criterion). *Let $G = (V, E)$ be a connected graph with cost function $c: E \rightarrow \mathbb{R}$, and let T be a spanning tree of G . Then T is a minimum spanning tree if and only if for every non-tree edge $f \in E \setminus E(T)$, the edge f has maximum cost among all edges in its fundamental cycle C_f .*

Proof. We prove both directions.

($\Rightarrow$): If T is a minimum spanning tree, then every non-tree edge has maximum cost in its fundamental cycle by [Lemma 2.1](#).

($\Leftarrow$): Suppose every non-tree edge has maximum cost in its fundamental cycle. We show T is a minimum spanning tree by proving $c(T) \leq c(T')$ for any spanning tree T' .

If $T = T'$, we are done. Otherwise, there exists an edge $e \in E(T) \setminus E(T')$. Write $e = \{s, t\}$ and let $X_e \subseteq V$ denote the component containing s after removing e from T , so that $t \in V \setminus X_e$. Since T' is connected, there exists a path P in T' from s to t . Because s and t lie on opposite sides of the partition $(X_e, V \setminus X_e)$, the path P must cross it at least once.

Pick any edge f on P that crosses the partition. Then f has one endpoint in X_e and the other in $V \setminus X_e$, hence $f \in \delta(X_e)$. Moreover, $f \in E(T')$ (since f lies on the path P) and $f \notin E(T)$ (since e is the unique tree edge in $\delta(X_e)$ by construction of the fundamental cut; see [Section 2.1](#)).

We construct a new spanning tree $T'' = (V, E(T') \setminus \{f\} \cup \{e\})$ by removing f and adding e . To verify that T'' is a spanning tree, note that f lies on the path P from s to t in T' , so removing f disconnects s from t , while adding $e = \{s, t\}$ reconnects them; thus T'' is connected. Since $|E(T'')| = |E(T')| - 1 + 1 = n - 1$, being connected with $n - 1$ edges makes T'' a spanning tree.

We now verify that $c(T'') \leq c(T')$. Since $f \notin E(T)$, the edge f creates a fundamental cycle C_f in T . The path in T connecting the endpoints of f must cross the partition $(X_e, V \setminus X_e)$, and because e is the unique tree edge in $\delta(X_e)$, this path contains e . Therefore $e \in C_f$. By assumption, f has maximum cost in C_f , so $c_f \geq c_e$. Hence

$$c(T'') = c(T') - c_f + c_e \leq c(T').$$

Moreover, $|E(T) \cap E(T'')| = |E(T) \cap E(T')| + 1$, since we added $e \in E(T)$ and removed $f \notin E(T)$.

Repeating this exchange, we obtain a sequence of spanning trees $T' = T_0, T_1, T_2, \dots$ where each T_{i+1} is derived from T_i by exchanging an edge of $E(T) \setminus E(T_i)$ with an edge of $E(T_i) \setminus E(T)$, satisfying $c(T_{i+1}) \leq c(T_i)$ and $|E(T) \cap E(T_{i+1})| = |E(T) \cap E(T_i)| + 1$. Since spanning trees have exactly $n - 1$ edges, the process terminates after at most $n - 1$ iterations with some $T_k = T$. The cost satisfies

$$c(T) = c(T_k) \leq c(T_{k-1}) \leq \dots \leq c(T_1) \leq c(T_0) = c(T').$$

Therefore $c(T) \leq c(T')$ for every spanning tree T' , which means T is a minimum spanning tree. $\square$

The cut-based criterion is analogous.

Theorem 2.2 (Cut Criterion). *Let $G = (V, E)$ be a connected graph with cost function $c: E \rightarrow \mathbb{R}$, and let T be a spanning tree of G . Then T is a minimum spanning tree if and only if for every tree edge $e \in E(T)$, the edge e has minimum cost among all edges in its fundamental cut $\delta(X_e)$.*

Proof. We prove both directions.

($\Rightarrow$): If T is a minimum spanning tree, then every tree edge has minimum cost in its fundamental cut by [Lemma 2.2](#).

($\Leftarrow$): Suppose every tree edge has minimum cost in its fundamental cut. We show $c(T) \leq c(T')$ for any spanning tree T' .

The argument mirrors the proof of [Theorem 2.1](#). If $T = T'$, we are done. Otherwise, pick any edge $e \in E(T) \setminus E(T')$ with $e = \{s, t\}$, and let X_e denote the component containing s in $T \setminus \{e\}$.

Since T' is connected, there exists a path P in T' from s to t . Because $s \in X_e$ and $t \in V \setminus X_e$ lie on opposite sides of the partition, the path P crosses $(X_e, V \setminus X_e)$ at least once. Let f be any edge on P crossing the partition. Then $f \in \delta(X_e)$, $f \in E(T')$, and $f \notin E(T)$ (since e is the unique tree edge in $\delta(X_e)$ and $f \neq e$). By the cut criterion, $c_e \leq c_f$.

By the same spanning tree verification as in [Theorem 2.1](#), the exchange $T'' = (V, E(T') \setminus \{f\} \cup \{e\})$ yields a spanning tree with

$$c(T'') = c(T') - c_f + c_e \leq c(T')$$

and $|E(T) \cap E(T'')| = |E(T) \cap E(T')| + 1$.

Repeating this exchange, we obtain a sequence $T' = T_0, T_1, \dots, T_k = T$ where each step satisfies $c(T_{i+1}) \leq c(T_i)$ and increases $|E(T) \cap E(T_i)|$ by one. The process terminates when $T_k = T$, yielding

$$c(T) = c(T_k) \leq c(T_{k-1}) \leq \dots \leq c(T_0) = c(T').$$

Therefore $c(T) \leq c(T')$ for every spanning tree T' , so T is a minimum spanning tree. $\square$

The two criteria are equivalent.

Theorem 2.3 (Equivalence of Criteria). *Let $G = (V, E)$ be a connected graph with cost function $c: E \rightarrow \mathbb{R}$, and let T be a spanning tree of G . Then T satisfies the cycle criterion if and only if T satisfies the cut criterion.*

Proof. By [Theorem 2.1](#), T satisfies the cycle criterion if and only if T is a minimum spanning tree. By [Theorem 2.2](#), T satisfies the cut criterion if and only if T is a minimum spanning tree. Therefore, T satisfies the cycle criterion if and only if T satisfies the cut criterion. $\square$

Together, these three theorems fully characterise minimum spanning tree optimality through local edge-level conditions, enabling verification without comparing against all exponentially many spanning trees. The characterisations extend naturally to the robust spanning tree problems in [Chapters 3 and 4](#), where similar exchange arguments establish optimality under uncertainty. The correctness of classical greedy algorithms such as Kruskal’s and Prim’s follows from these criteria, as discussed in [Section 2.3](#).

2.3 Kruskal and Prim: Algorithmic Remarks

The optimality criteria developed in [Section 2.2](#) directly yield the correctness of the two most classical MST algorithms: Kruskal’s algorithm [\[6\]](#) and Prim’s algorithm [\[7\]](#). Both are *greedy* methods that build a spanning tree incrementally, maintaining optimality conditions throughout [\[8, Section 2.3\]](#). A comprehensive account of minimum spanning tree algorithms is given by Ahuja, Magnanti and Orlin [\[9, Chapter 13\]](#). We describe each algorithm briefly, emphasising its connection to the criteria rather than implementation details.

Kruskal’s Algorithm. Kruskal’s algorithm constructs an MST by processing edges in non-decreasing order of cost. Starting with an empty edge set, the algorithm considers each edge in turn and adds it to the current forest if and only if it does not create a cycle. The process terminates when the forest becomes a spanning tree (after adding exactly $n - 1$ edges).

The correctness follows immediately from the cycle criterion ([Theorem 2.1](#)). When the algorithm terminates with spanning tree T , consider any non-tree edge $f \in E \setminus E(T)$. Since f was rejected, adding f would have created a cycle with edges already in T . All edges in this cycle were added before f was considered, hence each has cost at most c_f (by the sorting order). Therefore f has maximum cost in its fundamental cycle, satisfying the cycle criterion.

With an efficient union-find data structure to track connected components, Kruskal’s algorithm runs in $O(m \log n)$ time, dominated by the initial edge sorting [\[2, Theorem 6.5\]](#).

Prim’s Algorithm. Prim’s algorithm grows a single tree from an arbitrary starting vertex. At each step, it adds the minimum-cost edge crossing the cut between the current tree vertices and the remaining vertices. The process continues until all vertices are included.

The correctness follows by induction on the steps of the algorithm. Let S_t denote the set of tree vertices before step t and e_t the edge added at step t . We claim that the partial tree built so far can be extended to some MST of G at every step; since the algorithm terminates with a spanning tree, this implies the final tree is itself an MST.

The base case is trivial: the empty tree on the chosen root extends to any MST. For the inductive step, assume the partial tree T_{t-1} is contained in some MST T^* , and consider the edge e_t added next. If $e_t \in E(T^*)$, the induction continues with the same T^* .

Otherwise $e_t \notin E(T^*)$. The unique path in T^* between the endpoints of e_t starts in S_{t-1} (one endpoint of e_t lies there) and ends outside S_{t-1} (the other endpoint is the newly added vertex), so it must cross the cut $\delta(S_{t-1})$ at some edge $f \in E(T^*)$. By Prim's choice, e_t is the minimum-cost edge in $\delta(S_{t-1})$, hence $c_{e_t} \leq c_f$. The exchange $T^{**} = (T^* \setminus \{f\}) \cup \{e_t\}$ is again a spanning tree: adding e_t to T^* creates exactly its fundamental cycle, and f lies on that cycle, so removing f restores acyclicity. Its cost satisfies $c(T^{**}) = c(T^*) - c_f + c_{e_t} \leq c(T^*)$, so T^{**} is also an MST and contains $T_{t-1} \cup \{e_t\}$.

By induction, the spanning tree built by Prim's algorithm is an MST.

With a priority queue maintaining the minimum-cost edge to each non-tree vertex, Prim's algorithm achieves $O(m + n^2)$ time with a simple array implementation, or $O(m + n \log n)$ with Fibonacci heaps [2, Theorem 6.7].

Relevance to Robust Problems. Both algorithms solve the deterministic MST problem optimally in polynomial time. However, under uncertainty, the situation changes fundamentally. When edge costs are uncertain, neither algorithm directly applies: the greedy choices depend on which cost realisation occurs, yet the spanning tree must be chosen before costs are revealed. The robust spanning tree problems studied in Chapters 3 and 4 address precisely this challenge, and several variants become NP-hard despite the tractability of the deterministic case.

2.4 Uncertainty Models and Robust Optimisation

In many practical applications, edge costs are not known precisely at decision time. Construction expenses may fluctuate, travel times depend on traffic conditions, and communication link costs vary with demand. Robust optimisation addresses this challenge by seeking solutions that perform well across all possible cost realisations, rather than optimising for a single realisation fixed in advance [1]. This section formalises three uncertainty models and the corresponding robust objectives that we analyse in Chapters 3 and 4. All three require non-negative edge costs, reflecting the distances, construction expenses, and transmission delays that such costs typically represent, as is standard in the robust spanning tree literature [1, 5].

Discrete Scenarios. In the *discrete scenario model*, uncertainty is represented by a finite set of possible cost vectors.

Definition 2.1 (Discrete Uncertainty Set). A *discrete uncertainty set* consists of K explicitly given cost vectors:

$$\mathcal{U} = \{c^{(1)}, c^{(2)}, \dots, c^{(K)}\},$$

where each *scenario* $c^{(k)}$, for $k \in \{1, \dots, K\}$, is a cost vector in $\mathbb{R}_{\geq 0}^{|E|}$ assigning cost $c_e^{(k)}$ to every edge $e \in E$.

The parameter K denotes the number of scenarios. When K is a fixed constant (such as $K = 2$), we speak of a *constant number of scenarios*; when K is part of the input and may grow with the problem size, we speak of *unbounded K* . This distinction is crucial for computational complexity: several robust spanning tree

problems are weakly NP-hard already for a constant number of scenarios, admitting pseudo-polynomial algorithms and approximation schemes, but become strongly NP-hard when K is unbounded [10].

Interval Uncertainty. In the *interval model*, each edge cost may take any value within a specified range.

Definition 2.2 (Interval Uncertainty Set). *An interval uncertainty set is defined by lower and upper bounds on each edge:*

$$\mathcal{U} = \prod_{e \in E} [\ell_e, u_e] = \left\{ c \in \mathbb{R}_{\geq 0}^{|E|} : \ell_e \leq c_e \leq u_e \text{ for all } e \in E \right\},$$

where $0 \leq \ell_e \leq u_e$ for each edge $e \in E$.

Unlike the discrete model with finitely many scenarios, interval uncertainty encompasses a continuum of possible cost vectors. Despite this, many robust problems under interval uncertainty admit efficient solutions by exploiting the structure of the uncertainty set: worst-case costs are often attained at the boundary of the intervals. We develop these arguments in Chapters 3 and 4.

Budgeted Uncertainty. In the *budgeted model*, each edge has a baseline cost together with a bound on how far it may rise, and a single parameter caps how many edges may deviate from baseline at once.

Definition 2.3 (Budgeted Uncertainty Set). *Let $\bar{c} \in \mathbb{R}_{\geq 0}^{|E|}$ be a vector of nominal costs (the baseline value of each edge), $\hat{c} \in \mathbb{R}_{\geq 0}^{|E|}$ a vector of maximum deviations (the largest upward perturbation each edge can suffer), and $\Gamma \in [0, |E|]$ a deviation budget. The budgeted uncertainty set is*

$$\mathcal{U}^\Gamma = \left\{ c \in \mathbb{R}_{\geq 0}^{|E|} : c_e = \bar{c}_e + \hat{c}_e \delta_e, \delta_e \in [0, 1] \text{ for } e \in E, \sum_{e \in E} \delta_e \leq \Gamma \right\}.$$

Each edge cost c_e is determined by a deviation factor $\delta_e \in [0, 1]$: at $\delta_e = 0$ the edge takes its nominal cost $\bar{c}_e$, at $\delta_e = 1$ it takes the worst-case value $\bar{c}_e + \hat{c}_e$, and intermediate values interpolate linearly. The budget constraint $\sum_e \delta_e \leq \Gamma$ limits the cumulative deviation across the graph. Setting $\Gamma = 0$ recovers the deterministic problem on $\bar{c}$, while $\Gamma = |E|$ allows every edge to deviate independently and recovers an interval model with $[\ell_e, u_e] = [\bar{c}_e, \bar{c}_e + \hat{c}_e]$ [5]. The budget Γ thus offers a tunable middle ground: the modeller controls how conservative the analysis is through a single parameter, rather than committing in advance to a fixed scenario list or to the full interval ranges. As we shall see in Section 3.4, this added structure allows the min-max spanning tree problem to remain polynomial-time solvable under budgeted uncertainty.

Robust Objectives. Given an uncertainty set $\mathcal{U}$, we model robust optimisation as a sequential game: a decision-maker first selects a spanning tree T , then an adversary selects a worst-case scenario $c \in \mathcal{U}$. This structure motivates two robust objectives.

The *min-max objective* minimises the maximum cost over all scenarios:

$$\min_{T \in \mathcal{T}} \max_{c \in \mathcal{U}} c(T). \quad (2.2)$$

A spanning tree achieving this minimum is called a *min-max spanning tree*. This objective bounds the worst possible outcome regardless of which scenario materialises.

The *min-max regret objective* instead minimises the maximum *regret*, which measures how far a solution falls short of the scenario-specific optimum:

$$\text{Regret}(T, c) = c(T) - \text{MST}(c). \quad (2.3)$$

The min-max regret problem seeks a tree minimising the worst-case regret:

$$\min_{T \in \mathcal{T}} \max_{c \in \mathcal{U}} \text{Regret}(T, c) = \min_{T \in \mathcal{T}} \max_{c \in \mathcal{U}} [c(T) - \text{MST}(c)]. \quad (2.4)$$

A spanning tree achieving this minimum is called a *min-max regret spanning tree*. This objective minimises *opportunity cost*: the loss from not knowing the scenario in advance.

Computing regret requires knowing $\text{MST}(c)$, which depends on the scenario c . For discrete scenarios, we can precompute the MST for each scenario separately. For interval and budgeted uncertainty, the adversary's choice of c affects both $c(T)$ and $\text{MST}(c)$ simultaneously, and the worst-case scenario is generally not simply all upper bounds; identifying it requires the extremal arguments developed in [Chapters 3 and 4](#).

Worked Example: Discrete Scenarios on the Micro-graph. We illustrate the robust objectives using the discrete scenario model on the micro-graph from [Section 2.1](#). (Worked examples for interval and budgeted uncertainty require the extremal analysis developed in [Chapters 3 and 4](#); the budgeted case is treated in [Section 3.4](#).)

The micro-graph has vertices $V = \{1, 2, 3, 4\}$, edges $\{e_1, \dots, e_5\}$, and interval costs as specified in [Figure 2.2](#). We derive three discrete scenarios: $c^{(1)} = (2, 3, 1, 4, 5)$ (all lower bounds), $c^{(2)} = (4, 5, 7, 6, 7)$ (all upper bounds), and $c^{(3)} = (3, 4, 4, 5, 6)$ (interval midpoints). We evaluate three spanning trees: $T_1 = \{e_1, e_2, e_3\}$, $T_2 = \{e_1, e_2, e_4\}$, and $T_3 = \{e_2, e_3, e_5\}$.

[Table 2.1](#) records the cost and regret for each tree–scenario pair. The bottom row shows the MST cost for each scenario; note that the MST varies: T_1 is optimal under $c^{(1)}$ and $c^{(3)}$, whereas under $c^{(2)}$ the tree T_2 achieves the minimum cost 15.

To find the min-max regret tree, we proceed in two steps: compute each tree's maximum regret across scenarios (rightmost column), then select the tree with the smallest maximum. Tree T_1 has max regret 1, compared to 3 for T_2 and 4 for T_3 ; no other spanning tree of the micro-graph does better, so T_1 is the min-max regret spanning tree.

For the min-max objective, we compare worst-case costs: T_2 achieves 15 (under $c^{(2)}$), versus 16 for T_1 and 19 for T_3 . Again no other spanning tree improves on this, so T_2 is the min-max spanning tree. Here the two objectives select *different* trees: T_2 minimises worst-case cost, while T_1 minimises worst-case regret. This divergence illustrates that the choice of robust objective genuinely matters, a theme developed throughout [Chapters 3 and 4](#).

Table 2.1: Costs and regrets of three of the eight spanning trees of the micro-graph, under the three discrete scenarios $c^{(1)}$, $c^{(2)}$ and $c^{(3)}$ of Figure 2.2: the all-lower-bound, all-upper-bound and midpoint cost vectors. The final row gives the minimum spanning tree cost in each scenario, against which the regrets in the right-hand block are measured.

Tree	Cost $c^{(k)}(T)$			Regret			Max
	$k=1$	$k=2$	$k=3$	$k=1$	$k=2$	$k=3$	Regret
$T_1 = \{e_1, e_2, e_3\}$	6	16	11	0	1	0	1
$T_2 = \{e_1, e_2, e_4\}$	9	15	12	3	0	1	3
$T_3 = \{e_2, e_3, e_5\}$	9	19	14	3	4	3	4
MST($c^{(k)}$)	6	15	11				

2.5 Algorithmic Preliminaries

This section introduces the complexity and approximation terminology needed to interpret the results in Chapters 3 and 4. We focus on concepts directly relevant to robust spanning tree problems; for comprehensive treatments, see [5, Chapter 2] or [2, Chapter 15].

Polynomial Time and the Class P. An algorithm runs in *polynomial time* if, given an input of size n , it terminates in at most $O(n^k)$ steps for some fixed constant k . The class **P** consists of all problems solvable by a polynomial-time algorithm. Problems in **P** are considered tractable: their running time grows manageably with the input size. The minimum spanning tree problem lies in **P**, since Kruskal’s algorithm solves it in $O(m \log n)$ time (Section 2.3). Introducing uncertainty, however, can push problems outside **P**, as we shall see in Chapters 3 and 4.

Hard Problems: NP and NP-hardness. The class **NP** (nondeterministic polynomial time) contains problems whose candidate solutions can be *verified* in polynomial time, even if finding such a solution may itself be hard. To make this concrete, consider the decision version of MST: given a graph $G = (V, E)$, costs c , and a budget B , does there exist a spanning tree of cost at most B ? A candidate solution is a set $T \subseteq E$ of $n - 1$ edges. Verifying it requires two checks: that T is a spanning tree (one breadth-first search on the $n - 1$ edges of T runs in $O(n)$ time and confirms connectedness), and that the sum of its edge costs does not exceed B (another $O(n)$ additions). Since both checks are polynomial, the decision MST lies in **NP**.

A problem is *NP-hard* if every problem in **NP** can be *reduced* to it: transformed in polynomial time so that solving the target problem also solves the original. Informally, **NP-hard** problems are at least as hard as the hardest problems in **NP**. No polynomial-time algorithm is known for any **NP-hard** problem, and the widely believed conjecture $\mathbf{P} \neq \mathbf{NP}$ implies that none exists.

Weak and Strong NP-hardness. The knapsack problem is **NP-hard**, yet a textbook dynamic-programming algorithm solves it in $O(nW)$ time, where W is the

knapsack capacity. At first sight this seems to contradict $P \neq NP$. The resolution lies in how input size is measured: a number W is encoded in binary using only about $\log_2 W$ bits, not W itself. Doubling W therefore adds just one bit to the input but doubles the runtime, so $O(nW)$ is polynomial in the *value* of W but exponential in the *bit-length* of W , which is what “polynomial-time” strictly demands.

An algorithm is *pseudo-polynomial* if its running time is polynomial in the input size and in the magnitudes of the numeric values appearing in the input. Such an algorithm is not polynomial in the strict sense, but it is efficient whenever those values are not too large. This motivates two refinements of NP-hardness:

- A problem is *weakly NP-hard* if it is NP-hard and admits a pseudo-polynomial algorithm. The knapsack problem is the canonical example.
- A problem is *strongly NP-hard* if it is NP-hard even when all numeric values are bounded by a polynomial in n . Such problems admit no pseudo-polynomial algorithm unless $P = NP$.

This distinction is significant for robust spanning trees: the number of scenarios K plays a role analogous to a numeric input value, and several variants will turn out to be weakly NP-hard (Chapters 3 and 4).

Approximation Algorithms. When an exact polynomial-time algorithm is unlikely to exist, we settle for efficient algorithms returning provably near-optimal solutions.

An *α -approximation algorithm* for a minimisation problem is a polynomial-time algorithm that, on every instance, returns a solution of cost at most α times the optimum, where $\alpha \geq 1$ is the *approximation ratio*. The ratio is fixed for the algorithm: a 2-approximation guarantees a solution at most twice as costly as the optimum, with no tighter promise. Christofides’ algorithm for the metric travelling salesman problem, for instance, is a $\frac{3}{2}$ -approximation [2, Theorem 21.5].

For some problems, the ratio can instead be tuned by the user. A *polynomial-time approximation scheme* (PTAS) takes any $\varepsilon > 0$ as input and returns a $(1 + \varepsilon)$ -approximation in time polynomial in n for each fixed ε . Setting $\alpha := 1 + \varepsilon$, this is precisely an α -approximation for any $\alpha > 1$ chosen by the user, with smaller ε yielding tighter guarantees. The catch is that the runtime may grow rapidly in $1/\varepsilon$ (for instance, $O(n^{1/\varepsilon})$), so a PTAS need not be practical for very small ε .

A *fully polynomial-time approximation scheme* (FPTAS) is a PTAS whose running time is additionally polynomial in $1/\varepsilon$. The trade-off then becomes graceful: halving ε at most polynomially increases the runtime. The knapsack problem admits an FPTAS [2, Theorem 17.9], illustrating that even some NP-hard problems can be approximated to within any desired precision in practical time. However, strongly NP-hard problems cannot have an FPTAS unless $P = NP$.

Relevance to This Thesis. The complexity of robust spanning tree problems hinges on two factors:

1. **Number of scenarios K :** For discrete uncertainty, several problems are already weakly NP-hard for a constant number of scenarios, admitting pseudo-polynomial algorithms and approximation schemes, and become strongly NP-hard when K is part of the input.

2. **Uncertainty model:** For interval uncertainty, the continuum of scenarios prevents enumeration. Some variants remain in $\mathbf{P}$ by exploiting MST structure (Section 2.2); others are $\mathbf{NP}$ -hard.

Chapters 3 and 4 classify each problem variant accordingly and present polynomial-time algorithms or approximations as appropriate.

Summary

Two technical pillars established in this chapter recur throughout the rest of the thesis. The first is the local characterisation of MST optimality through fundamental cycles and cuts (Theorems 2.1 and 2.2): the exchange arguments built on these criteria will reappear repeatedly in the robust setting. The second is the family of uncertainty models (discrete scenarios, interval uncertainty, and budgeted uncertainty) combined with robust objectives (min-max cost and min-max regret), which yields the problem variants analysed in Chapters 3 and 4. Several of these variants will turn out to be $\mathbf{NP}$ -hard, in sharp contrast to the polynomial-time tractability of the deterministic MST.

Chapter 3

Min-Max Spanning Tree

Chapter 2 established that the deterministic minimum spanning tree problem, in which all edge costs are known in advance, is solvable in $O(m \log n)$ time. This chapter studies the min-max version (2.2), where the costs are not known when the tree is chosen. A single spanning tree must be committed to in advance. An adversary then selects the least favourable cost realisation from a prescribed uncertainty set, and the tree is judged by its cost under that worst case.

Section 3.1 writes this objective out at the edge level and examines its structure under each of the three uncertainty models from Section 2.4, isolating the inner worst-case evaluator that drives the subsequent analysis. Section 3.2 treats interval uncertainty, where the problem reduces to a minimum spanning tree under the upper-bound cost vector and is therefore polynomial. Section 3.3 treats discrete uncertainty: weakly NP-hard already at $K = 2$, admitting an FPTAS for any constant K , and strongly NP-hard when K grows with the input, though still approximable within a logarithmic factor in K . Section 3.4 treats budgeted uncertainty, polynomial via a richer reformulation, and a closing summary draws the three models together.

3.1 Problem Formulation

The min-max objective (2.2) introduced in Section 2.4 is stated with the cost function $c(T)$ left unexpanded. We now write it out at the edge level using $c(T) = \sum_{e \in E(T)} c_e$. The resulting *min-max spanning tree problem* is

$$\min_{T \in \mathcal{T}} \max_{c \in \mathcal{U}} \sum_{e \in E(T)} c_e. \quad (3.1)$$

The uncertainty set $\mathcal{U}$ takes one of three forms in this thesis (Definitions 2.1 to 2.3), and the algorithmic character of the problem varies sharply with the choice.

Decision Version and Complexity Classification. The optimisation problem (3.1) asks for both the optimal value and a tree achieving it. The associated *decision version* asks, given a budget $B \in \mathbb{R}$, whether there exists a spanning tree $T \in \mathcal{T}$ with $\max_{c \in \mathcal{U}} c(T) \leq B$. This decision version lies in NP whenever the inner maximisation $\max_{c \in \mathcal{U}} c(T)$ is polynomial-time computable for a fixed tree: the certificate is the tree T , and verification requires confirming $T \in \mathcal{T}$ (an $O(n)$ breadth-first search, as in the deterministic MST decision problem of Section 2.5) and then

evaluating the inner maximum. Under discrete uncertainty this evaluation is the explicit maximum $\max_{k \in \{1, \dots, K\}} c^{(k)}(T)$ over the K scenarios and takes $O(Kn)$ time; under interval and budgeted uncertainty, polynomial inner evaluation follows from the extremal characterisations developed in [Sections 3.2](#) and [3.4](#). The complexity terminology used in the remainder of the chapter (polynomial time, weak and strong NP-hardness, FPTAS) is the one fixed in [Section 2.5](#).

Worst-Case Evaluator. A central object throughout the chapter is the *worst-case cost* of a fixed spanning tree $T \in \mathcal{T}$:

$$\text{wc}(T) := \max_{c \in \mathcal{U}} \sum_{e \in E(T)} c_e. \quad (3.2)$$

With this notation the min-max problem [\(3.1\)](#) reads simply $\min_{T \in \mathcal{T}} \text{wc}(T)$: among all spanning trees, choose the one whose worst-case cost is smallest. How difficult this outer minimisation is depends on the form that $\text{wc}(T)$ takes as a function of the tree T .

The most favourable situation is when $\text{wc}(T)$ is a sum of fixed per-edge weights, that is,

$$\text{wc}(T) = \sum_{e \in E(T)} w_e$$

for some weight vector w that does not itself depend on the choice of T . An objective of this form is called *linear*, and minimising a linear objective over all spanning trees is nothing other than a deterministic minimum spanning tree problem under the weights w , solvable by Kruskal’s algorithm in $O(m \log n)$ time. The three uncertainty models differ precisely in whether $\text{wc}(T)$ is linear in this sense.

Under *interval* uncertainty it is. [Lemma 3.1](#) will show that the adversary does best by setting every edge of T to its upper bound, so that $\text{wc}(T) = \sum_{e \in E(T)} u_e$. This is a sum of fixed per-edge weights with $w = u$, and the min-max problem is therefore a minimum spanning tree problem under the upper-bound cost vector u .

Under *discrete* uncertainty it is not. Here $\text{wc}(T) = \max_{k \in \{1, \dots, K\}} c^{(k)}(T)$ is the largest of the K scenario costs of T , that is, the pointwise maximum of the K linear functions $c^{(1)}(T), \dots, c^{(K)}(T)$. Whereas the interval adversary chooses each c_e independently and is captured once and for all by the fixed vector $w = u$, the discrete adversary instead commits to one full scenario covering all edges, and which scenario is worst can depend on the tree T . In general no fixed weight vector w reproduces this tree-dependent choice, so $\text{wc}(T)$ need not collapse into a single sum $\sum_{e \in E(T)} w_e$ across all spanning trees. Consequently the problem does not reduce to one MST computation, and this obstruction is the source of the hardness established in [Section 3.3](#).

Under *budgeted* uncertainty $\text{wc}(T)$ is in general not linear either, because the shared deviation budget couples the edges of T : which of them the adversary drives upward depends on the tree. The problem therefore does not reduce to a single MST computation either, but [Section 3.4](#) shows that it nonetheless remains polynomial.

3.2 Interval Worst-Case Characterisation

We begin with interval uncertainty, the model identified in [Section 3.1](#) as the favourable case. There the worst-case cost $\text{wc}(T)$ requires maximising over the interval set $\mathcal{U} = \prod_{e \in E} [\ell_e, u_e]$, which contains a continuum of cost vectors. The following lemma shows that this maximisation has a transparent solution: confronted with a fixed tree, the adversary simply charges every edge its upper bound.

Lemma 3.1 (Interval Extremal Cost). *Let $\mathcal{U} = \prod_{e \in E} [\ell_e, u_e]$ be an interval uncertainty set. For every spanning tree $T \in \mathcal{T}$, the worst-case cost satisfies*

$$\text{wc}(T) = \max_{c \in \mathcal{U}} \sum_{e \in E(T)} c_e = \sum_{e \in E(T)} u_e,$$

and the maximum is attained at the cost vector c^* defined by $c_e^* = u_e$ for every edge $e \in E$.

Proof. Fix a spanning tree $T \in \mathcal{T}$. Only the edges of T appear in the sum $\sum_{e \in E(T)} c_e$, so the costs assigned to edges outside T do not affect the value of the maximisation. Because the uncertainty set $\mathcal{U} = \prod_{e \in E} [\ell_e, u_e]$ is a Cartesian product, the cost of each edge can be chosen from its interval independently of the others. Since each variable c_e appears in exactly one term of the sum, the maximum of the sum equals the sum of the per-edge maxima:

$$\max_{c \in \mathcal{U}} \sum_{e \in E(T)} c_e = \sum_{e \in E(T)} \max_{c_e \in [\ell_e, u_e]} c_e.$$

Each cost c_e ranges over the interval $[\ell_e, u_e]$ and is maximised at u_e . Substituting yields $\max_{c \in \mathcal{U}} \sum_{e \in E(T)} c_e = \sum_{e \in E(T)} u_e$. The cost vector c^* with $c_e^* = u_e$ for all $e \in E$ lies in $\mathcal{U}$ and attains this value; its entries on edges outside T are immaterial. $\square$

[Lemma 3.1](#) reduces the worst-case cost of a tree to a single sum of upper bounds; this is the linear form anticipated in [Section 3.1](#), with weight vector $w = u$. The consequence for the min-max problem is immediate.

Corollary 3.1 (Polynomial Solvability under Interval Uncertainty). *Under interval uncertainty $\mathcal{U} = \prod_{e \in E} [\ell_e, u_e]$, the min-max spanning tree problem satisfies*

$$\min_{T \in \mathcal{T}} \text{wc}(T) = \min_{T \in \mathcal{T}} \sum_{e \in E(T)} u_e = \text{MST}(u),$$

where $\text{MST}(u)$ is the minimum spanning tree cost under the upper-bound cost vector $u = (u_e)_{e \in E}$. The problem is solved by computing a minimum spanning tree under u , which Kruskal's algorithm performs in $O(m \log n)$ time; in particular, it lies in $\mathbf{P}$.

Proof. By [Lemma 3.1](#), $\text{wc}(T) = \sum_{e \in E(T)} u_e$ for every $T \in \mathcal{T}$. Minimising over T gives $\min_{T \in \mathcal{T}} \text{wc}(T) = \min_{T \in \mathcal{T}} \sum_{e \in E(T)} u_e$, which is the minimum spanning tree cost $\text{MST}(u)$ by (2.1). Kruskal's algorithm computes a minimum spanning tree under any fixed cost vector in $O(m \log n)$ time ([Section 2.3](#)); applied to u , it returns an optimal min-max tree. $\square$

The reduction in [Corollary 3.1](#) is the spanning tree instance of [5, Theorem 4.8], which establishes it for interval min-max problems in general.

Worked Example. We apply [Lemma 3.1](#) to the micro-graph of [Figure 2.2](#), whose upper-bound cost vector is $u = (4, 5, 7, 6, 7)$. By the lemma, the worst-case cost of a tree is the sum of its edges' upper bounds, so for the three representative trees

$$\text{wc}(T_1) = 4 + 5 + 7 = 16, \quad \text{wc}(T_2) = 4 + 5 + 6 = 15, \quad \text{wc}(T_3) = 5 + 7 + 7 = 19.$$

These values coincide with the scenario- $c^{(2)}$ column of [Table 2.1](#), as they must: the micro-graph's scenario $c^{(2)}$ was defined as the all-upper-bounds vector, so $c^{(2)} = u$.

To find the min-max spanning tree, [Corollary 3.1](#) directs us to a minimum spanning tree under u . Kruskal's algorithm processes the edges in non-decreasing order of u , namely e_1 (4), e_2 (5), e_4 (6), followed by e_3 and e_5 (both 7). It adds $e_1 = \{1, 2\}$, $e_2 = \{2, 3\}$ and $e_4 = \{3, 4\}$; the tree is then complete, and the two remaining edges are not examined. The result is $T_2 = \{e_1, e_2, e_4\}$ with cost $\text{MST}(u) = 15$. The three trees evaluated above are only three of the micro-graph's eight spanning trees; what certifies T_2 as the min-max tree over all of them is [Corollary 3.1](#), applied through this minimum spanning tree computation, rather than the three-way comparison alone. The entries of u are not all distinct, e_3 and e_5 sharing the upper bound 7, so the sufficient condition of [Remark 2.1](#) does not apply here. The minimum is nonetheless attained by T_2 alone: the seven remaining spanning trees have u -costs between 16 and 20. Hence T_2 is the unique min-max spanning tree, its cost matching the value $\text{wc}(T_2) = 15$ computed above.

The Role of the Lower Bounds. [Lemma 3.1](#) shows that the min-max objective depends on the interval data only through the upper bounds u ; the lower bounds ℓ play no role, and the entire interval model compresses to the single cost vector u . Two points are worth drawing out. First, the separation step in the proof used only that $\mathcal{U}$ is a product of intervals; it is exactly this absence of coupling between edges that fails under budgeted uncertainty ([Section 3.4](#)), where a shared deviation budget links the edges and the worst case is in general no longer a fixed per-edge sum. Second, the insensitivity to the lower bounds is special to the min-max objective: the regret objective of [Chapter 4](#) measures each tree against the scenario-optimal tree, a comparison that does depend on the lower bounds, and there the interval problem becomes substantially harder.

3.3 Complexity and Approximation

Under interval uncertainty the entire model compressed into the single cost vector u , and the min-max problem collapsed to one minimum spanning tree computation ([Lemma 3.1](#) and [Corollary 3.1](#)). Discrete uncertainty admits no such compression: by (3.2), the worst-case cost is the largest of the K scenario costs of the tree, and the maximising scenario may change from tree to tree ([Section 3.1](#)). This section turns that obstruction into complexity results, and the price depends on whether the number of scenarios is a fixed constant or part of the input. For every constant K the problem is weakly NP-hard, already for $K = 2$; this is [Theorem 3.1](#), the hardness result of this chapter that we prove in full, complemented by a pseudo-polynomial algorithm and an FPTAS ([Theorems 3.2](#) and [3.3](#)). When K is part of the input, the hardness becomes strong and no constant-factor approximation exists at all, the best guarantee being an $O(\log K)$ -approximation; these results close the section.

3.3.1 Discrete Scenarios: Constant K

Throughout this subsection the number of scenarios K is a constant, fixed independently of the instance.

Theorem 3.1 (Hardness for Two Scenarios). *Under discrete uncertainty, the min-max spanning tree problem is weakly NP-hard, even when restricted to $K = 2$ scenarios.*

[Theorem 3.1](#) is due to Kouvelis and Yu [1]; the presentation here follows Goerigk and Hartisch [5, Theorem 8.4]. The reduction starts from the [partition problem](#), which is NP-complete [2, Corollary 15.28]: given positive integers $w_1, \dots, w_n$, decide whether some subset $P \subseteq \{1, \dots, n\}$ satisfies $\sum_{i \in P} w_i = \frac{1}{2} \sum_{i=1}^n w_i$. Throughout, $W = \sum_{i=1}^n w_i$ denotes the total weight and $Q = W/2$ the target value.

From a partition instance we build the grid graph $G = (V, E)$ shown in [Figure 3.1](#), with two rows and $n + 1$ columns. Its vertex set is

$$V = \{v_{i,j} : i \in \{1, 2\}, j \in \{0, 1, \dots, n\}\},$$

and two vertices are adjacent exactly when they agree in one index and differ by one in the other, that is,

$$\{v_{i,j}, v_{k,\ell}\} \in E \iff (i = k \text{ and } |j - \ell| = 1) \text{ or } (|i - k| = 1 \text{ and } j = \ell).$$

For $j \in \{1, \dots, n\}$ we write $a_j = \{v_{1,j-1}, v_{1,j}\}$ for the *first-row edge of column j* and $b_j = \{v_{2,j-1}, v_{2,j}\}$ for the *second-row edge of column j* ; together these $2n$ edges are the *horizontal edges*, each named after the column of its right endpoint. The remaining $n + 1$ edges are the *vertical edges* $r_j = \{v_{1,j}, v_{2,j}\}$, one for every column $j \in \{0, 1, \dots, n\}$. The two scenarios place each weight once in each row:

$$c_{a_j}^{(1)} = c_{b_j}^{(2)} = w_j, \quad c_{b_j}^{(1)} = c_{a_j}^{(2)} = 0 \quad \text{for } j \in \{1, \dots, n\},$$

and $c_{r_j}^{(1)} = c_{r_j}^{(2)} = 0$ for all $j \in \{0, 1, \dots, n\}$.

The intended mechanism is visible in [Figure 3.1](#). A spanning tree must pass from column $j - 1$ to column j through a_j or b_j , paying w_j in scenario 1 or in scenario 2 respectively, so choosing a tree distributes the weights $w_1, \dots, w_n$ between the two scenarios; the worst case is the larger of the two shares, smallest when the split is even. The proof makes this correspondence exact.

Proof of Theorem 3.1. The construction is computable in polynomial time: G has $2(n + 1)$ vertices and $3n + 1$ edges, and every cost is zero or one of the given weights. We claim that G admits a spanning tree of worst-case cost at most Q if and only if the partition instance is a yes-instance; this is the decision version of [Section 3.1](#) with budget $B = Q$.

Step 1: some optimal tree contains every vertical edge. Suppose a spanning tree T does not contain the vertical edge r_j of some column j . Adding r_j to T creates exactly one cycle (property [P3](#)), the fundamental cycle C_{r_j} , consisting of r_j and the unique T -path from $v_{1,j}$ to $v_{2,j}$. This path connects the two rows, so it crosses between them along some vertical edge of T ; as $r_j \notin E(T)$, that edge is some r_ℓ

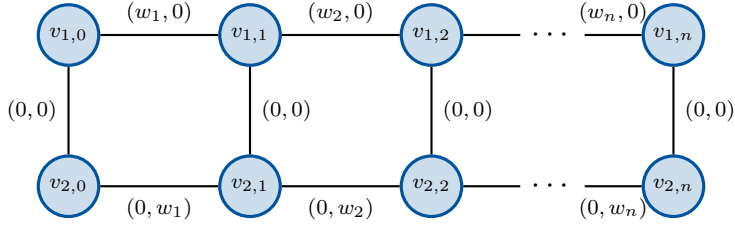


Figure 3.1: The grid graph G built from a partition instance with weights $w_1, \dots, w_n$. Each edge is labelled with its cost pair $(c_e^{(1)}, c_e^{(2)})$: a first-row edge of column j costs w_j in scenario 1, a second-row edge of column j costs w_j in scenario 2, and every vertical edge costs zero in both scenarios.

with $\ell \neq j$, and in travelling between columns j and ℓ the path uses at least one horizontal edge h . The exchange $T' = (T \setminus \{h\}) \cup \{r_j\}$ is again a spanning tree: deleting h breaks the unique cycle of $T \cup \{r_j\}$, while the remainder of C_{r_j} keeps the endpoints of h connected. Since $c_{r_j}^{(k)} = 0 \leq c_h^{(k)}$ for $k \in \{1, 2\}$, neither scenario cost increases, and $\text{wc}(T') \leq \text{wc}(T)$. Each exchange raises the number of vertical edges by one, so iterating turns any spanning tree into one that contains all vertical edges at no larger worst-case cost; in particular, the minimum worst-case cost is attained by such a tree.

Step 2: with the vertical edges fixed, a spanning tree is just a choice of one row per column, and its worst-case cost depends on that choice alone. With every vertical edge present, the two vertices of each column are already joined, and a spanning tree's only remaining decision is, for each column j , whether it crosses from column $j - 1$ through the first-row edge a_j or the second-row edge b_j . For a subset $P \subseteq \{1, \dots, n\}$, let T_P consist of every vertical edge together with the first-row edge a_j in the columns $j \in P$ and the second-row edge b_j in the rest:

$$T_P := \{r_0, r_1, \dots, r_n\} \cup \{a_j : j \in P\} \cup \{b_j : j \notin P\}.$$

We show that the trees containing every vertical edge are exactly the sets T_P . Each T_P is a spanning tree: the vertical edges join the two rows within every column and the n chosen horizontal edges link consecutive columns, so T_P is connected and, with its $(n + 1) + n = |V| - 1$ edges, is a tree. Conversely, let T be a spanning tree containing every vertical edge, and fix $j \in \{1, \dots, n\}$. Let $V_{<j} := \{v_{i,\ell} : i \in \{1, 2\}, \ell \in \{0, 1, \dots, j - 1\}\}$ be the vertices of the first j columns. The cut $\delta(V_{<j}) = \{a_j, b_j\}$ consists of the only two edges straddling the boundary between columns $j - 1$ and j (Figure 3.1), so connectivity forces at least one of a_j, b_j into $E(T)$, while $r_{j-1}, r_j \in E(T)$ forbid both, since together they would close the four-cycle a_j, r_j, b_j, r_{j-1} . Thus T contains exactly one edge of each pair $\{a_j, b_j\}$, and as G has only vertical and horizontal edges, $T = T_P$ with $P = \{j : a_j \in E(T)\}$. The worst-case cost now follows: scenario 1 charges only first-row edges and scenario 2 only second-row edges, so with $w(P) = \sum_{j \in P} w_j$,

$$c^{(1)}(T_P) = w(P), \quad c^{(2)}(T_P) = W - w(P), \quad \text{wc}(T_P) = \max\{w(P), W - w(P)\}.$$

Step 3: a tree of worst-case cost at most Q exists if and only if the weights split evenly. By Steps 1 and 2 the minimisation over trees becomes a minimisation over subsets,

$$\min_{T \in \mathcal{T}} \text{wc}(T) = \min_{P \subseteq \{1, \dots, n\}} \max\{w(P), W - w(P)\}.$$

Since $w(P)$ and $W - w(P)$ sum to $2Q$, their maximum is at least Q , with equality if and only if $w(P) = Q$. The minimum worst-case cost is therefore at least Q , and equals Q exactly when some subset satisfies $w(P) = Q$, which is the partition condition. As the minimum never falls below Q , a spanning tree of worst-case cost at most Q exists if and only if the partition instance is a yes-instance, proving the claim.

The partition problem therefore reduces in polynomial time to the decision version of the min-max spanning tree problem with $K = 2$, and the problem is **NP**-hard. Since [Theorem 3.2](#) provides a pseudo-polynomial algorithm, the hardness is weak in the sense of [Section 2.5](#). $\square$

The restriction to $K = 2$ is the sharpest form of the result rather than a limitation: for any constant $K \geq 2$, duplicating one of the two scenarios $K - 2$ times changes no worst-case cost, so the problem with exactly K scenarios is weakly **NP**-hard as well.

A Small Instance. Consider the weights $w = (3, 1, 2)$, so $n = 3$, $W = 6$ and $Q = 3$. The subset $P = \{1\}$ satisfies $w(P) = 3 = Q$, and the tree from the proof,

$$T_P = \{r_0, r_1, r_2, r_3\} \cup \{a_1, b_2, b_3\},$$

has $c^{(1)}(T_P) = w_1 = 3$ and $c^{(2)}(T_P) = w_2 + w_3 = 3$, hence $\text{wc}(T_P) = 3 = Q$; by Step 3 no spanning tree does better, and the reduction reports a yes-instance. The weights $w = (1, 5)$ give the same $W = 6$ and $Q = 3$, but the four subsets $P \subseteq \{1, 2\}$ yield $w(P) \in \{0, 1, 5, 6\}$ and worst-case costs $\max\{w(P), W - w(P)\} \in \{5, 6\}$; the optimum is $5 > Q$, so no spanning tree meets the threshold, matching the fact that no subset of $\{1, 5\}$ sums to 3.

[Theorem 3.1](#) proves the problem hard, but its reduction reveals the hardness to be of a mild and familiar kind. All of the difficulty came from splitting a set of weights into two equal halves, the partition problem. None came from the spanning tree structure, which merely carried the two scenarios. Hardness of this numerical kind is weak: it bites only when the numbers are large and eases once they are kept small. The knapsack problem of [Section 2.5](#) is the canonical example, **NP**-hard yet admitting an FPTAS. For a constant number of scenarios, the min-max spanning tree problem follows the same pattern.

Theorem 3.2 (Pseudo-Polynomial Algorithm). *For every constant K , the min-max spanning tree problem under discrete uncertainty is solvable in pseudo-polynomial time.*

Theorem 3.3 (FPTAS). *For every constant K , the min-max spanning tree problem under discrete uncertainty admits a fully polynomial-time approximation scheme.*

Both algorithms are built on one subroutine, the *exact spanning tree problem*: given integer edge costs and a target value, does some spanning tree have exactly that total cost? This is solvable in pseudo-polynomial time. Its running time grows with the magnitude of the costs, not with the size of the graph alone [11].

The subroutine turns the min-max problem into a search. Under K scenarios a tree carries a list of K costs, one per scenario, and its worst-case cost is the largest entry of that list. The min-max optimum is the tree whose largest entry is smallest. Each of the K entries is an integer between zero and the total cost of all edges in its own scenario, so when K is fixed only pseudo-polynomially many lists can occur. Searching these few lists replaces the search over the exponentially many spanning trees. Bundling each edge's K costs into a single number, in a base wide enough that the scenarios cannot interfere, lets the subroutine test whether a prescribed list is realised by some tree. Iterating it over the candidate lists and keeping the realised one of smallest maximum returns the min-max optimum in pseudo-polynomial time [5, Theorem 8.2].

The approximation scheme runs the same search on scaled costs. Dividing every cost by a common factor, chosen from ε and a bound on the optimum, and then rounding shrinks the magnitudes until only polynomially many lists remain. This turns the pseudo-polynomial search into a polynomial one, while distorting each scenario cost by at most a factor $1 + \varepsilon$. The returned tree therefore has worst-case cost within $1 + \varepsilon$ of the optimum, in time polynomial in the input and in $1/\varepsilon$ [5, Theorem 8.3]. This scheme is due to Aissi, Bazgan and Vanderpooten [12].

These results supply the pseudo-polynomial algorithm that Theorem 3.1's proof invoked to classify its hardness as weak. With an FPTAS available as well, the problem cannot be strongly hard unless $P = NP$, since a strongly hard problem would admit neither. Both guarantees hold only because K is fixed, which keeps the number of cost lists small. This is exactly what the next subsection removes: once K grows with the input, the lists are no longer few, and both the algorithm and the approximation scheme fall away.

3.3.2 Discrete Scenarios: Unbounded K

With K part of the input, the hardness of the problem changes in kind. For a fixed number of scenarios the difficulty was numerical, the partition problem in disguise, and that is why it was weak and yielded to scaling. With arbitrarily many scenarios available, a purely logical problem can be encoded instead, one that involves no large numbers at all, and the problem becomes strongly hard and far harder to approximate.

Theorem 3.4 (Strong NP-Hardness and Inapproximability). *If K is part of the input, the min-max spanning tree problem under discrete uncertainty is strongly NP-hard, and unless $P = NP$ it cannot be approximated within a factor of $2 - \varepsilon$ for any $\varepsilon > 0$.*

The result is due to Kasperski and Zieliński [13]; the presentation here follows Goerigk and Hartisch [5, Theorem 8.5]. The reduction starts from 3-SAT, which is NP-complete [2, Theorem 15.22]: given a Boolean formula whose clauses each contain three literals, each a variable or its negation, decide whether some truth assignment satisfies every clause.

The graph receives a small gadget for each clause, shown in Figure 3.2, and the gadgets are chained one after another, so that every route from one end of the graph to the other must pass through each gadget in turn. Inside the gadget of clause i lie three parallel two-edge routes from a_i to b_i ; the first edge $\{a_i, v_{i,j}\}$ of the j -th route represents the j -th literal $L_{i,j}$ of the clause, while the second edge, like the edges linking consecutive gadgets, costs zero in every scenario. Crossing the gadget requires at least one of its literal edges, so every spanning tree selects at least one literal in every clause, and the only costs a tree can ever incur sit on the literal edges it selects. The scenarios then encode the logic: for every complementary pair of occurrences, $L_{i,j} = \neg L_{i',j'}$, a scenario is added that charges cost 1 to the two edges $\{a_i, v_{i,j}\}$ and $\{a_{i'}, v_{i',j'}\}$ and 0 to every other edge.

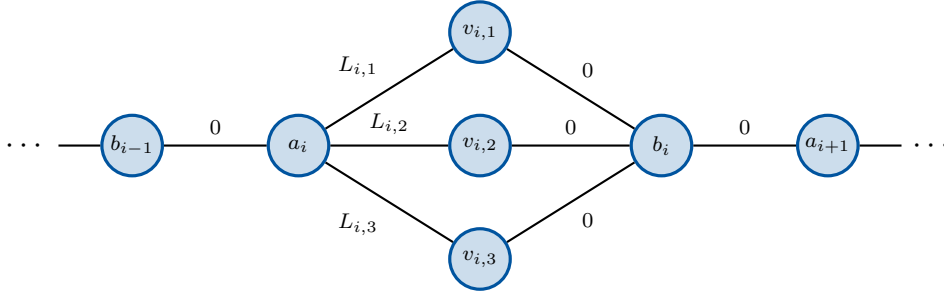


Figure 3.2: The gadget of clause i in the reduction from 3-SAT, chained to its neighbouring gadgets through b_{i-1} and a_{i+1} . The edge $\{a_i, v_{i,j}\}$ carries the j -th literal $L_{i,j}$ of the clause; a scenario charges cost 1 to the two literal edges of a complementary pair of occurrences and 0 everywhere else. All edges labelled 0 are free in every scenario, so a tree's cost arises only on the literal edges it selects, and crossing the gadget from a_i to b_i forces at least one of them.

The correspondence is exact in both directions. If the formula is satisfiable, fix a satisfying assignment and build the tree that takes, in each clause, one literal edge whose literal the assignment makes true, together with all the cost-free edges. Each gadget contributes five vertices and four selected edges, with one connector joining each consecutive pair, so the selection has one edge fewer than the graph has vertices; being connected as well, it is a spanning tree. No two selected literals are complementary, both being true under one assignment, so every scenario charges at most one selected edge, and the worst-case cost is at most 1. Conversely, suppose some spanning tree avoids both edges of every complementary pair. Setting each of its selected literals to true is then consistent, and since every clause contributes a selected literal, the formula is satisfied. For an unsatisfiable formula every spanning tree therefore contains both edges of some pair, and the scenario of that pair costs it 2; since no scenario has more than two unit-cost edges, no tree costs more than 2 in any scenario, and the optimal worst-case cost is exactly 2.

The optimal worst-case cost is thus at most 1 for a satisfiable formula and exactly 2 for an unsatisfiable one, so deciding which of the two occurs decides 3-SAT, and the problem is NP-hard. Every cost in the construction is 0 or 1, so the numbers remain polynomially bounded and the hardness is strong, in contrast to the partition reduction, where large weights were essential.

The same gap yields the inapproximability bound. Suppose that, for some $\varepsilon > 0$, a polynomial-time algorithm always returned a spanning tree of worst-case cost at most $2 - \varepsilon$ times the optimum. On a satisfiable instance the optimum is at most 1, so the returned tree would cost at most $2 - \varepsilon < 2$; on an unsatisfiable instance every tree costs exactly 2, the returned one included. Reading off whether the returned tree costs less than 2 would therefore decide 3-SAT in polynomial time, which is impossible unless $\mathbf{P} = \mathbf{NP}$. The inapproximability is in fact far stronger, no constant factor being achievable at all [14]; the self-contained $2 - \varepsilon$ bound above is its modest form.

Theorem 3.5 (Logarithmic Approximation). *The min-max spanning tree problem under discrete uncertainty can be approximated within a factor of $O(\log K)$.*

This closes the gap from above, leaving only a logarithmic distance to the no-constant-factor barrier. The obstacle is the worst-case objective itself: a maximum over the K scenario costs is far harder to minimise than the plain sum that reduced the interval problem to a single spanning tree computation (Corollary 3.1). The remedy is to minimise a sum after all, one built to imitate the maximum. For a parameter $p \geq 1$, aggregate the K entries of a tree's cost list into the p -th root of the sum of their p -th powers, in the literature the p -norm of the list. Since the scenario costs are non-negative (Definition 2.1), the aggregate brackets the worst case,

$$\text{wc}(T) \leq \left(\sum_{k=1}^K (c^{(k)}(T))^p \right)^{1/p} \leq K^{1/p} \text{wc}(T), \quad (3.3)$$

the first inequality because the sum contains the p -th power of its largest entry, the second because it consists of K terms, none larger. The larger p is, the closer the factor $K^{1/p}$ draws to one and the more faithfully the aggregate tracks the maximum.

Unlike the maximum, the aggregate is an objective a greedy method can handle. Growing the tree edge by edge in the manner of Kruskal's algorithm, but adding at each step, among the edges creating no cycle, one that increases the aggregate the least, returns a spanning tree whose aggregate is at most $p/\ln 2$ times the smallest possible [5, Theorem 5.21]. The two error sources now pull against each other: raising p tightens the right-hand inequality of (3.3) but loosens the greedy guarantee. Both are paid in succession. By the left inequality of (3.3) the greedy tree's worst-case cost is at most its aggregate, and that aggregate is at most $p/\ln 2$ times the smallest aggregate any spanning tree attains, in particular at most $p/\ln 2$ times the aggregate of a min-max optimal tree. By the right inequality, that last aggregate is at most $K^{1/p}$ times the min-max optimum. The greedy tree therefore lies within a factor $(p/\ln 2) \cdot K^{1/p}$ of that optimum. Choosing $p = \ln K$ makes both factors explicit: writing $K = e^{\ln K}$ gives $K^{1/p} = K^{1/\ln K} = e$, while $p/\ln 2 = \ln K/\ln 2 = \log_2 K$, so that

$$\frac{p}{\ln 2} \cdot K^{1/p} = e \log_2 K = O(\log K),$$

which is the guarantee of Theorem 3.5 [5, Corollary 5.22]. The bound is due to Baak et al. [15], building on the greedy analysis of Bilò et al. [16].

The discrete-uncertainty picture is now complete, and the number of scenarios is its dividing line. For a fixed number of scenarios the problem is only weakly hard and

admits an FPTAS; once that number is part of the input, it becomes strongly hard, approximable within $O(\log K)$ but within no constant factor. One problem thus carries two kinds of hardness: a numerical one that scaling defeats, and a structural one that it cannot touch.

3.4 Budgeted Uncertainty

The last of the three uncertainty models restores polynomial-time solvability, though by a longer route than interval uncertainty required. In the budgeted set $\mathcal{U}^\Gamma$ of Definition 2.3, each edge may rise from its nominal cost $\bar{c}_e$ by at most its maximum deviation $\hat{c}_e$, but a single budget caps how much deviation the adversary may distribute in total, $\sum_{e \in E} \delta_e \leq \Gamma$. That shared budget couples the edges of the chosen tree, as Section 3.1 observed. The maximisation therefore no longer separates edge by edge as it did in Section 3.2, because budget spent on one edge is budget another cannot have, and which edges the adversary drives upward depends on the tree. In general no fixed weight vector reproduces $\text{wc}(T)$, and the collapse to a single minimum spanning tree that settled the interval case (Corollary 3.1) is unavailable.

What replaces it is a richer reformulation. The adversary's tree-dependent choice can be absorbed into a single parameter, though this means solving not one deterministic minimum spanning tree problem but a small family of them, one for each candidate value of that parameter. As in Section 3.2, we begin with a fixed tree and turn to the minimisation over all trees afterwards.

Worst-Case Cost of a Fixed Tree. Fix a spanning tree $T \in \mathcal{T}$. Evaluating (3.2) over $\mathcal{U}^\Gamma$ confronts the adversary with a choice its interval counterpart never faced: unless the budget covers the whole tree, it cannot raise every edge of T at once and must decide where a limited allowance does the most damage. The choice is nevertheless an easy one, namely to spend the budget on the edges of T carrying the largest deviations, raising each of them in full. The following lemma confirms that this greedy rule is optimal and evaluates the worst case in closed form.

Lemma 3.2 (Budgeted Extremal Cost). *Let $\mathcal{U}^\Gamma$ be a budgeted uncertainty set with nominal costs $\bar{c}$, maximum deviations $\hat{c}$, and an integer deviation budget $\Gamma \geq 0$. Let $T \in \mathcal{T}$ be a spanning tree, order its edges as $E(T) = \{f_1, \dots, f_{n-1}\}$ so that $\hat{c}_{f_1} \geq \hat{c}_{f_2} \geq \dots \geq \hat{c}_{f_{n-1}}$, and set $g := \min\{\Gamma, n-1\}$. Then the worst-case cost of T satisfies*

$$\text{wc}(T) = \sum_{e \in E(T)} \bar{c}_e + \sum_{j=1}^g \hat{c}_{f_j},$$

and the maximum in (3.2) is attained at the cost vector c^* given by $c_{f_j}^* = \bar{c}_{f_j} + \hat{c}_{f_j}$ for $j \leq g$ and $c_e^* = \bar{c}_e$ for every other edge $e \in E$.

Proof. Every $c \in \mathcal{U}^\Gamma$ has the form $c_e = \bar{c}_e + \hat{c}_e \delta_e$ with $\delta \in [0, 1]^{|E|}$ and $\sum_{e \in E} \delta_e \leq \Gamma$ (Definition 2.3). The nominal part of $c(T)$ is the same for every scenario, so only the deviations are at stake and (3.2) becomes

$$\text{wc}(T) = \sum_{e \in E(T)} \bar{c}_e + \max \left\{ \sum_{e \in E(T)} \hat{c}_e \delta_e : \delta \in [0, 1]^{|E|}, \sum_{e \in E} \delta_e \leq \Gamma \right\}. \quad (3.4)$$

It suffices to show that this maximum equals $\sum_{j=1}^g \hat{c}_{f_j}$ and is attained at the deviation vector realising c^* .

The objective in (3.4) involves only the edges of T . Setting $\delta_e := 0$ for every $e \notin E(T)$ therefore leaves the objective unchanged while freeing budget, so some maximiser vanishes outside $E(T)$, and for such δ the budget constraint reads $\sum_{j=1}^{n-1} \delta_{f_j} \leq \Gamma$.

If $g = 0$, then either $\Gamma = 0$, so that the budget forces every δ_e to zero, or T has no edges at all; in both cases the maximum is the empty sum, as claimed. Assume from now on that $g \geq 1$.

The candidate maximiser is the vector δ^* with $\delta_{f_j}^* = 1$ for $j \leq g$ and $\delta_e^* = 0$ for every other edge, which realises exactly the cost vector c^* . Its total deviation is $g \leq \Gamma$, so δ^* is feasible, and its objective value is $\sum_{j=1}^g \hat{c}_{f_j}$. The maximum is therefore at least this value, and it remains to show that no feasible δ exceeds it.

Let δ be feasible and vanishing outside $E(T)$. Comparing its value with that of δ^* and grouping the terms according to whether $j \leq g$,

$$\sum_{j=1}^{n-1} \hat{c}_{f_j} \delta_{f_j} - \sum_{j=1}^g \hat{c}_{f_j} = \sum_{j=1}^g \hat{c}_{f_j} (\delta_{f_j} - 1) + \sum_{j=g+1}^{n-1} \hat{c}_{f_j} \delta_{f_j}.$$

We bound the right-hand side by replacing every coefficient $\hat{c}_{f_j}$ with the single value $\hat{c}_{f_g}$, the smallest deviation that δ^* pays for. By the chosen ordering this replacement can only increase the terms, in each sum for its own reason. For $j \leq g$ the factor $\delta_{f_j} - 1$ is non-positive while $\hat{c}_{f_j} \geq \hat{c}_{f_g}$, and scaling a non-positive quantity by a smaller coefficient increases it. For $j > g$ the factor δ_{f_j} is non-negative while $\hat{c}_{f_j} \leq \hat{c}_{f_g}$, and scaling a non-negative quantity by a larger coefficient increases it. Both sums then carry $\hat{c}_{f_g}$ as a common factor, so

$$\sum_{j=1}^{n-1} \hat{c}_{f_j} \delta_{f_j} - \sum_{j=1}^g \hat{c}_{f_j} \leq \hat{c}_{f_g} \left(\sum_{j=1}^{n-1} \delta_{f_j} - g \right).$$

The bracket is never positive. If $\Gamma \leq n - 1$, then $g = \Gamma$ and the budget constraint gives $\sum_{j=1}^{n-1} \delta_{f_j} \leq \Gamma = g$; if $\Gamma > n - 1$, then $g = n - 1$ and the bounds $\delta_{f_j} \leq 1$ alone give $\sum_{j=1}^{n-1} \delta_{f_j} \leq n - 1 = g$. Since $\hat{c}_{f_g} \geq 0$ by Definition 2.3, the whole right-hand side is at most zero, so δ does not beat δ^* . The maximum in (3.4) is therefore $\sum_{j=1}^g \hat{c}_{f_j}$, attained at δ^* . $\square$

The lemma assumes an integer budget, whereas Definition 2.3 allows any $\Gamma \in [0, |E|]$. A fractional budget changes little: the adversary fills the largest deviations in the same order, and any budget left over once whole edges are exhausted is spent on the next largest deviation, which adds at most one fractional term to the closed form. Every budget appearing in what follows is an integer.

Evaluating $\text{wc}(T)$ is now a sorting task. Ordering the $n - 1$ deviations of $E(T)$ and adding the g largest to the nominal cost of T takes $O(n \log n)$ time, which is the polynomial inner evaluation promised for the budgeted model in Section 3.1.

The formula also displays both ends of the budget range. At $\Gamma = 0$ it returns $\text{wc}(T) = \sum_{e \in E(T)} \bar{c}_e$, so the min-max problem is the deterministic minimum spanning tree problem under the nominal costs $\bar{c}$. Once $\Gamma \geq n - 1$ we have $g = n - 1$, every edge of T deviates fully, and $\text{wc}(T) = \sum_{e \in E(T)} (\bar{c}_e + \hat{c}_e)$ is the interval worst case of

[Lemma 3.1](#) on the box $\prod_{e \in E} [\bar{c}_e, \bar{c}_e + \hat{c}_e]$. [Definition 2.3](#) reaches that box at $\Gamma = |E|$, but against spanning trees the interval behaviour sets in earlier, at $\Gamma = n - 1$: a tree offers the adversary only $n - 1$ edges to spend on, and any further budget is wasted. Across the whole range $\text{wc}(T)$ is non-decreasing in Γ , since enlarging the budget enlarges $\mathcal{U}^\Gamma$. At both ends it is once more a sum of fixed per-edge weights, with $w = \bar{c}$ and $w = \bar{c} + \hat{c}$ respectively. Only for budgets strictly between them can the coupling described above take effect.

Minimisation over All Trees. [Lemma 3.2](#) selects deviations by rank, and rank is what stands in the way of minimising over all trees. Whether an edge belongs to the g largest is a question about the other edges of its tree as much as about the edge itself, so the selection moves with T and cannot be written down edge by edge.

Rank can be traded for a threshold. Fix a level $\pi \geq 0$ and charge each edge only the part of its deviation that exceeds it, namely $[\hat{c}_e - \pi]^+$ with $[z]^+ := \max\{z, 0\}$, a quantity the edge settles on its own. An edge whose deviation reaches the level is charged exactly π less than that deviation, so a flat payment of $\Gamma\pi$ can make good Γ such shortfalls at once. Placing the level at the Γ -th largest deviation of T , for $1 \leq \Gamma \leq n - 1$, balances the account: the Γ largest deviations all reach that level and one copy of π restores each of them in full, while the remaining edges do not exceed it and are charged nothing. The account then totals exactly the worst-case cost of [Lemma 3.2](#), and the following lemma shows that no level does better.

Lemma 3.3 (Threshold Form of the Worst-Case Cost). *Let $\mathcal{U}^\Gamma$ be a budgeted uncertainty set with an integer deviation budget $\Gamma \geq 0$. Every spanning tree $T \in \mathcal{T}$ then satisfies*

$$\text{wc}(T) = \min_{\pi \geq 0} \left[\Gamma\pi + \sum_{e \in E(T)} \left(\bar{c}_e + [\hat{c}_e - \pi]^+ \right) \right],$$

and the minimum is attained at some $\pi \in \{0\} \cup \{\hat{c}_e : e \in E(T)\}$.

Proof. Order $E(T) = \{f_1, \dots, f_{n-1}\}$ by non-increasing deviations as in [Lemma 3.2](#), and put $g := \min\{\Gamma, n - 1\}$. The nominal costs do not involve π and move outside the minimisation, so by [Lemma 3.2](#) the claim is equivalent to

$$\min_{\pi \geq 0} \varphi_T(\pi) = \sum_{j=1}^g \hat{c}_{f_j} \quad \text{for} \quad \varphi_T(\pi) := \Gamma\pi + \sum_{j=1}^{n-1} [\hat{c}_{f_j} - \pi]^+,$$

with a minimiser among 0 and the deviations of T . We treat three ranges of the budget; in each we exhibit a level of the required form that attains the claimed value, and then verify that no level falls below it.

Suppose first that $\Gamma = 0$, so that $g = 0$ and the claimed value is the empty sum. If T has no edges, then φ_T vanishes identically; otherwise the level $\pi = \hat{c}_{f_1}$ makes every bracket vanish, giving $\varphi_T(\hat{c}_{f_1}) = 0$. No level falls below this, because every summand of φ_T is non-negative.

Suppose next that $1 \leq \Gamma \leq n - 1$, so that $g = \Gamma$. At the level $\pi = \hat{c}_{f_g}$ the ordering gives $[\hat{c}_{f_j} - \hat{c}_{f_g}]^+ = \hat{c}_{f_j} - \hat{c}_{f_g}$ for $j < g$ and $[\hat{c}_{f_j} - \hat{c}_{f_g}]^+ = 0$ for $j \geq g$, so that

$$\varphi_T(\hat{c}_{f_g}) = g \hat{c}_{f_g} + \sum_{j=1}^{g-1} (\hat{c}_{f_j} - \hat{c}_{f_g}) = \sum_{j=1}^g \hat{c}_{f_j},$$

the flat payment restoring each of the g largest deviations to its full value. No level falls below this value either, as two elementary bounds show. Let $\pi \geq 0$. The summands of φ_T with $j > g$ are non-negative and may be dropped, and each remaining bracket satisfies $[z]^+ \geq z$, so

$$\varphi_T(\pi) \geq \Gamma\pi + \sum_{j=1}^g [\hat{c}_{f_j} - \pi]^+ \geq \Gamma\pi + \sum_{j=1}^g (\hat{c}_{f_j} - \pi) = \sum_{j=1}^g \hat{c}_{f_j},$$

the final equality because $\Gamma = g$, so that the g subtracted copies of π cancel the flat payment.

Suppose finally that $\Gamma > n - 1$, so that $g = n - 1$. At the level $\pi = 0$ every bracket returns its deviation unchanged, giving $\varphi_T(0) = \sum_{j=1}^{n-1} \hat{c}_{f_j}$. For any $\pi \geq 0$, applying $[z]^+ \geq z$ to all $n - 1$ summands,

$$\varphi_T(\pi) \geq \Gamma\pi + \sum_{j=1}^{n-1} (\hat{c}_{f_j} - \pi) = \sum_{j=1}^{n-1} \hat{c}_{f_j} + (\Gamma - (n - 1))\pi,$$

and the final term is non-negative because $\Gamma > n - 1$ and $\pi \geq 0$.

The three ranges partition the integer budgets $\Gamma \geq 0$, and in each the exhibited level lies in $\{0\} \cup \{\hat{c}_e : e \in E(T)\}$, which proves the lemma. $\square$

The attaining level need not be unique: at $\Gamma = n - 1$ the minimum is attained both at $\pi = \hat{c}_{f_{n-1}}$ and at $\pi = 0$, where the brackets already return every deviation in full. For the minimisation over all trees this does not matter; what matters is that some minimiser lies in a set fixed in advance. The candidate set of [Lemma 3.3](#) still depends on T , but enlarging it to the deviations of all edges makes it the same for every tree, and the two minimisations can then be interchanged.

Theorem 3.6 (Polynomial Solvability under Budgeted Uncertainty). *Let $\mathcal{U}^\Gamma$ be a budgeted uncertainty set with an integer deviation budget $\Gamma \geq 0$, put $\Pi := \{0\} \cup \{\hat{c}_e : e \in E\}$, and for $\pi \in \Pi$ let c^π be the cost vector with $c_e^\pi := \bar{c}_e + [\hat{c}_e - \pi]^+$. Then the min-max spanning tree problem under budgeted uncertainty satisfies*

$$\min_{T \in \mathcal{T}} \text{wc}(T) = \min_{\pi \in \Pi} \left[\Gamma\pi + \text{MST}(c^\pi) \right],$$

and if π^ attains the right-hand minimum, then every minimum spanning tree under c^{π^*} is min-max optimal. The problem is solvable in $O(m^2 \log n)$ time; in particular, it lies in $\mathbf{P}$.*

The result is due to Bertsimas and Sim [\[3\]](#), who prove it for min-max problems over an arbitrary set of binary solutions; the presentation here follows Goerigk and Hartisch [\[5, Theorem 4.21\]](#), in whose derivation the threshold form of [Lemma 3.3](#) appears as an intermediate reformulation.

Proof. For every $T \in \mathcal{T}$ the set Π contains $\{0\} \cup \{\hat{c}_e : e \in E(T)\}$ and therefore, by [Lemma 3.3](#), a minimiser of the threshold form; enlarging the candidate set in this way

is what removes its dependence on T [5, Lemma 4.20]. Restricting a minimisation to a subset containing a minimiser leaves its value unchanged, so

$$\text{wc}(T) = \min_{\pi \in \Pi} \left[\Gamma\pi + \sum_{e \in E(T)} c_e^\pi \right] \quad \text{for every } T \in \mathcal{T}.$$

Minimising over $\mathcal{T}$ and interchanging the two minimisations, both of which run over finite sets, gives

$$\min_{T \in \mathcal{T}} \text{wc}(T) = \min_{\pi \in \Pi} \min_{T \in \mathcal{T}} \left[\Gamma\pi + \sum_{e \in E(T)} c_e^\pi \right] = \min_{\pi \in \Pi} \left[\Gamma\pi + \min_{T \in \mathcal{T}} \sum_{e \in E(T)} c_e^\pi \right],$$

the second step because $\Gamma\pi$ does not depend on the tree. At fixed π the vector c^π is one cost vector for all trees alike, so the inner problem is the deterministic minimum spanning tree problem (2.1) under c^π , of value $\text{MST}(c^\pi)$. This proves the identity.

For the second claim, let π^* attain the outer minimum and let T^* be a minimum spanning tree under c^{π^*} . Then

$$\min_{T \in \mathcal{T}} \text{wc}(T) = \Gamma\pi^* + \sum_{e \in E(T^*)} c_e^{\pi^*} \geq \min_{\pi \in \Pi} \left[\Gamma\pi + \sum_{e \in E(T^*)} c_e^\pi \right] = \text{wc}(T^*) \geq \min_{T \in \mathcal{T}} \text{wc}(T),$$

so equality holds throughout and T^* is min-max optimal.

It remains to bound the running time. The set Π holds at most $m + 1$ levels, one deviation per edge together with zero. A single level costs $O(m \log n)$ time: assembling c^π needs one subtraction per edge, and Kruskal's algorithm then returns $\text{MST}(c^\pi)$ in $O(m \log n)$ time (Section 2.3). Sweeping the levels and retaining the smallest value therefore takes $O(m^2 \log n)$ time in total. $\square$

Under interval uncertainty a single minimum spanning tree computation settled the problem (Corollary 3.1); under a budget, at most $m + 1$ of them do. The adversary's choice of where to spend, which changes from tree to tree, has been reduced to a single level, and levels are few enough to enumerate.

Worked Example. The micro-graph of Figure 2.2 shows both how the enumeration of Theorem 3.6 runs and what the budget buys as it grows. We take nominal costs at the interval midpoints and maximum deviations at the half-widths,

$$\bar{c} = (3, 4, 4, 5, 6), \quad \hat{c} = (1, 1, 3, 1, 1),$$

so that $\bar{c}$ coincides entry by entry with the midpoint scenario $c^{(3)}$ of Table 2.1, while $\bar{c} + \hat{c} = (4, 5, 7, 6, 7)$ agrees with the upper-bound scenario $c^{(2)}$, the vector u of Section 3.2. Neither coincidence belongs to the budgeted model, which is fixed by $\bar{c}$, $\hat{c}$ and Γ alone; both follow from how we have instantiated it here. They are convenient nonetheless, because the two ends of the budget range will return values that Table 2.1 has already recorded. We first work through one budget completely and then let the budget vary.

Fix the budget $\Gamma = 2$. [Lemma 3.2](#) evaluates each tree directly: the worst-case cost of a tree is its nominal cost plus its two largest deviations, so for the three representative trees

$$\text{wc}(T_1) = 11 + 3 + 1 = 15, \quad \text{wc}(T_2) = 12 + 1 + 1 = 14, \quad \text{wc}(T_3) = 14 + 3 + 1 = 18.$$

The remaining five spanning trees have worst-case costs between 16 and 19, so T_2 is the unique min-max tree, at 14.

[Theorem 3.6](#) reaches the same tree without evaluating any tree individually. Its candidate levels are

$$\Pi = \{0\} \cup \{\hat{c}_e : e \in E\} = \{0, 1, 3\},$$

three rather than the six that $m + 1$ permits, since the repeated deviations collapse to one candidate. [Table 3.1](#) performs the three deterministic solves. Its outer rows hold two cost vectors the chapter has met before: at $\pi = 0$ nothing is discounted and $c^0 = u$, while at $\pi = 3$ every deviation is absorbed and $c^3 = \bar{c}$. The middle row wins: the level $\pi^* = 1$ gives the total $12 + 2 \cdot 1 = 14$, with minimum spanning tree T_2 . The two routes agree, in the value and in the tree.

Table 3.1: The reformulation of [Theorem 3.6](#) on the micro-graph at $\Gamma = 2$. Each row solves one deterministic minimum spanning tree problem under the modified costs c^π ; the smallest total identifies the optimal value and an optimal tree.

π	$c^\pi = \bar{c} + [\hat{c} - \pi]^+$	$\text{MST}(c^\pi)$	$\Gamma\pi$	Total
0	(4, 5, 7, 6, 7)	15	0	15
1	(3, 4, 6, 5, 6)	12	2	14
3	(3, 4, 4, 5, 6)	11	6	17

One budget exercises the machinery; the range of budgets shows the model. [Figure 3.3](#) records the optimal value at every integer budget, and we read it from left to right. [Table 3.1](#) keeps working throughout: only its flat-payment column depends on Γ , so the same three rows answer each budget that follows.

At the left end, $\Gamma = 0$, the optimum is 11, attained by T_1 : the deterministic minimum spanning tree under $\bar{c}$, whose value appears in [Table 2.1](#) as $\text{MST}(c^{(3)})$. At this budget the flat payments vanish, so the $\text{MST}(c^\pi)$ column alone decides: its smallest entry, 11 at the level $\pi = 3$, is the optimum. The level thus moves opposite to the budget here: with nothing to spend, the cheapest account is the one that absorbs every deviation. The level is not a budget in disguise, and reading it as one inverts its role.

One unit of budget already moves the tree. At $\Gamma = 1$ the optimum is 13, attained by T_2 , while T_1 falls to second place: $\text{wc}(T_1) = 11 + 3 = 14$ against $\text{wc}(T_2) = 12 + 1 = 13$. The reason is e_3 , an edge of T_1 whose deviation 3 is the largest in the graph. T_1 is therefore the cheapest tree when no deviation can occur, and among the most exposed as soon as one can occur; the optimum switches to T_2 at $\Gamma = 1$ and stays there.

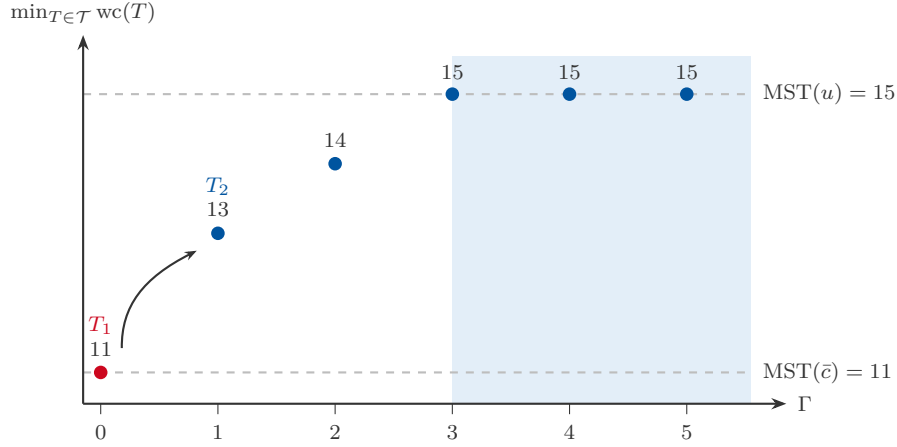


Figure 3.3: Optimal budgeted min-max value on the micro-graph at each integer budget Γ . The dashed lines mark the deterministic optimum under $\bar{c}$ and the interval optimum under u ; the shaded region is $\Gamma \geq n - 1$, where every tree’s worst case equals its interval worst case. The red point is the only budget at which T_1 is optimal; from $\Gamma = 1$ onwards the optimum is attained by T_2 (arrow: the switch).

At the right end the value saturates. From $\Gamma = 3$ onwards the optimum is 15, which is $MST(u)$ from [Section 3.2](#): every tree’s worst case now equals its interval worst case. Saturation arrives at $\Gamma = n - 1 = 3$, well before the $\Gamma = |E| = 5$ at which [Definition 2.3](#) reaches the full box, because a spanning tree of this graph offers only three edges on which to spend. Exactly at the saturation budget the attaining level is no longer unique, $\pi = 0$ and $\pi = 1$ both yielding 15, the tie anticipated after [Lemma 3.3](#); beyond it, nothing changes.

Taken together, the three readings show the budget acting as a dial between the deterministic problem and the interval model. Its ends reproduce two optima already on record, in [Table 2.1](#) and [Section 3.2](#), while in between the value climbs and, more tellingly, the optimal tree changes: a single parameter carries the problem from one model to the other and alters the decision on the way.

Summary

The three uncertainty models of this chapter are separated by a single question: how far the adversary’s best answer depends on the tree it faces. Under interval uncertainty it does not depend on the tree at all. [Lemma 3.1](#) identifies one cost vector, the upper bounds, that is simultaneously worst for every tree, so the worst-case cost is a sum of fixed per-edge weights and the outer minimisation is a single deterministic minimum spanning tree computation ([Corollary 3.1](#)).

Under discrete uncertainty the dependence is genuine. A tree that is safe in one scenario may be exposed in another, so which scenario witnesses the worst case is a property of the tree, and in general no fixed vector can stand in for the scenario set. The trade-off this forces is what the reductions of [Section 3.3](#) exploit, and two scenarios already suffice to make the problem weakly NP-hard.

Under budgeted uncertainty the answer depends on the tree as well, since the budget is spent on the chosen tree’s own largest deviations, and yet the problem

remains polynomial. The reason is that a single scalar removes the dependence: once a level is fixed, every tree is costed edge by edge again, and at most $m + 1$ levels can matter ([Theorem 3.6](#)).

The dividing line is therefore not linearity of the worst-case cost, which two of the three models lack in general. It is whether the dependence on the tree can be stripped away by a bounded amount of extra work: none is needed under intervals, one enumerated parameter suffices under a budget, and under discrete scenarios no such device can exist unless $P = NP$.

The extremal characterisations, the two-scenario reduction and the budgeted reformulation are proved in full here. The strong-hardness construction and the logarithmic approximation are developed from their sources in the text. Only the pseudo-polynomial algorithm and the approximation scheme for constant K are stated on the authority of the literature. Two of the three models return in [Chapter 4](#) under the regret objective, where the ordering changes: the interval case, the straightforward one here, becomes the hard one.

Chapter 4

Min-Max Regret Spanning Tree

Chapter 3 judged a spanning tree by an absolute standard, the cost it incurs in the worst case, with no other tree entering the verdict. This chapter keeps the setting: a tree is still committed before the costs are revealed, and an adversary still selects from a prescribed uncertainty set. What changes is the standard. The min-max regret objective (2.4) charges the tree only the gap between its cost and the best cost available in the same scenario, so the adversary now works to widen a gap rather than to inflate a total.

Section 4.1 writes this objective out at the edge level and isolates the evaluator the chapter minimises. Section 4.2 treats interval uncertainty: an extremal characterisation as explicit as the min-max one exists, the problem is strongly NP-hard nonetheless, and a 2-approximation is available. Section 4.3 treats discrete uncertainty, where the hierarchy of Section 3.3 returns: weakly NP-hard already at $K = 2$, pseudo-polynomial and approximable for any constant K , and strongly NP-hard once K is part of the input. Budgeted uncertainty is treated in this thesis only for the min-max objective; Section 5.1 records by citation what is known for its regret counterpart. A closing summary draws the two models together.

4.1 Regret Formulation

The regret (2.3) introduced in Section 2.4 measures a spanning tree $T \in \mathcal{T}$ under a realised cost vector c by the difference $\text{Regret}(T, c) = c(T) - \text{MST}(c)$. Expanding the tree cost edge by edge turns the min-max regret objective (2.4) into the *min-max regret spanning tree problem*

$$\min_{T \in \mathcal{T}} \max_{c \in \mathcal{U}} \left[\sum_{e \in E(T)} c_e - \text{MST}(c) \right]. \quad (4.1)$$

The bracket subtracts an optimal value from a tree cost. Only its first term depends on the tree; the second, $\text{MST}(c)$, is a floor set by the scenario alone, and no spanning tree costs less. A scenario that is expensive for T therefore gains the adversary nothing if it raises the floor by just as much.

The floor also shifts every tree equally. Under a single fixed scenario, regret and cost differ by the constant $\text{MST}(c)$ and therefore rank the spanning trees identically, so the two objectives can differ only through the maximisation. On the micro-graph

they do differ: across the three scenarios recorded in [Table 2.1](#), the min-max tree is T_2 and the min-max regret tree is T_1 .

Decision Version and Complexity Classification. The decision version asks, for a bound $B \in \mathbb{R}$, whether some spanning tree has maximum regret at most B ; the tree itself is the certificate, as in [Section 3.1](#), so membership in **NP** turns on evaluating the inner maximum in polynomial time. That evaluation is costlier than its min-max counterpart, because every scenario is now costed against its own optimum. Under discrete uncertainty the maximum regret of T is the largest of the K differences $c^{(k)}(T) - \text{MST}(c^{(k)})$, and evaluating it from scratch takes $O(Km \log n)$ time against $O(Kn)$ for the worst-case cost. The surcharge is paid once rather than per tree: the K minimum spanning trees do not depend on T , so after computing them a single time each further tree costs $O(Kn)$, as under the min-max objective. Under interval uncertainty the evaluation rests on the extremal characterisation of [Section 4.2.1](#).

Maximum-Regret Evaluator. The quantity the chapter minimises is the *maximum regret* of a fixed spanning tree $T \in \mathcal{T}$,

$$\text{wcr}(T) := \max_{c \in \mathcal{U}} \text{Regret}(T, c) = \max_{c \in \mathcal{U}} \left[\sum_{e \in E(T)} c_e - \text{MST}(c) \right], \quad (4.2)$$

the counterpart of the worst-case cost (3.2). The problem in (4.1) collapses to $\min_{T \in \mathcal{T}} \text{wcr}(T)$, and the chapter turns on two questions about $\text{wcr}(T)$: what form does it take as a function of the tree, and can that form be minimised over all spanning trees at once?

For the worst-case cost, one answer settles both questions. Over the interval set, [Lemma 3.1](#) lets every cost climb to its upper bound independently of the rest, so the evaluator becomes a sum of fixed per-edge weights, and minimising such a sum is a single deterministic minimum spanning tree computation. No such reduction is available for the maximum regret. The floor is the reason: $\text{MST}(c)$ depends on all m coordinates of c at once, so the bracket in (4.2) is not a sum of independent per-edge terms, and its maximum is not the sum of separate per-edge maxima. [Section 2.4](#) already records the consequence: the all-upper-bounds vector, which is the worst case under the min-max objective, is in general not the worst case for regret. Locating the maximum requires an argument that treats both terms of the bracket together.

These two questions organise the two sections that follow. Under discrete uncertainty the first is immediate, the maximum in (4.2) running over the K scenarios of a finite uncertainty set, so [Section 4.3](#) is occupied entirely by the second. Under interval uncertainty even the first requires proof.

4.2 Interval Regret

[Section 4.2.1](#) answers the first of the two questions and [Section 4.2.2](#) the second. The answers point in opposite directions: the maximum regret of a fixed tree is as easy to write down as the worst-case cost was in [Section 3.2](#), and far harder to minimise.

4.2.1 Extremal Characterisation

The scenario that answers the first question can be written down before anything is proved. The difficulty identified in [Section 4.1](#) is that raising the cost of T gains the adversary nothing if the floor rises with it. On an interval set it resolves edge by edge. An edge of T should cost the most its interval allows: T pays for it, and the scenario-optimal tree need not. An edge outside T should cost the least its interval allows: T never pays for it, and the scenario-optimal tree may. The two rules touch disjoint sets of edges, so they can be followed together, and following both yields one scenario per tree. What the rules do not settle is that nothing better exists, precisely because the first can lift the floor as it raises the cost of T ; settling that is the proof's work.

Lemma 4.1 (Interval Extremal Regret). *Let $\mathcal{U} = \prod_{e \in E} [\ell_e, u_e]$ be an interval uncertainty set. For every spanning tree $T \in \mathcal{T}$, the maximum regret satisfies*

$$\text{wcr}(T) = \max_{c \in \mathcal{U}} \text{Regret}(T, c) = \text{Regret}(T, c^T) = \sum_{e \in E(T)} u_e - \text{MST}(c^T),$$

where the maximum is attained at the cost vector $c^T \in \mathcal{U}$ defined by $c_e^T = u_e$ for $e \in E(T)$ and $c_e^T = \ell_e$ for $e \in E \setminus E(T)$.

Proof. We show that no scenario in $\mathcal{U}$ gives T a larger regret than c^T does, comparing T inside each scenario with a minimum spanning tree of that scenario.

Let $c \in \mathcal{U}$ and let $S \in \mathcal{T}$ be a minimum spanning tree under c , so that $\text{MST}(c) = c(S)$ and $\text{Regret}(T, c) = c(T) - c(S)$. Then

$$\begin{aligned} \text{Regret}(T, c) &= c(T) - c(S) = \sum_{e \in E(T) \setminus E(S)} c_e - \sum_{e \in E(S) \setminus E(T)} c_e \\ &\leq \sum_{e \in E(T) \setminus E(S)} u_e - \sum_{e \in E(S) \setminus E(T)} \ell_e \\ &= \sum_{e \in E(T) \setminus E(S)} c_e^T - \sum_{e \in E(S) \setminus E(T)} c_e^T \\ &= c^T(T) - c^T(S) \\ &\leq c^T(T) - \text{MST}(c^T) = \text{Regret}(T, c^T). \end{aligned}$$

The first equality holds because every edge of $E(T) \cap E(S)$ contributes the same amount to $c(T)$ and to $c(S)$. The inequality replaces each cost of the first sum by its upper bound and each cost of the second, subtracted, sum by its lower bound; both moves are available because $c \in \mathcal{U}$ confines every c_e to $[\ell_e, u_e]$. The second equality is the definition of c^T : the first index set lies inside $E(T)$, where c^T takes upper bounds, and the second inside $E \setminus E(T)$, where it takes lower bounds. The third equality returns the shared edges, again at identical cost to both trees. The final inequality holds because S is a spanning tree, so its cost under c^T is at least $\text{MST}(c^T)$, and the resulting expression is $\text{Regret}(T, c^T)$ by definition of the regret.

The chain holds for every $c \in \mathcal{U}$, and c^T itself lies in $\mathcal{U}$, since each of its entries is an endpoint of the corresponding interval, so the bound is attained. The maximum regret of T is therefore $\text{Regret}(T, c^T)$, and it equals $\sum_{e \in E(T)} u_e - \text{MST}(c^T)$ because c^T sets every edge of T to its upper bound. $\square$

[Lemma 4.1](#) answers the first of [Section 4.1](#)’s two questions: as a function of the tree, the maximum regret is a sum of upper bounds less a minimum spanning tree cost, both taken at a scenario read off the tree itself. The characterisation is the spanning tree instance of [\[5, Theorem 4.12\]](#), which establishes it for interval min-max regret problems in general, following Aissi, Bazgan and Vanderpooten [\[10\]](#). The evaluative consequence is immediate.

Corollary 4.1 (Polynomial Evaluation under Interval Uncertainty). *Under interval uncertainty, the maximum regret $\text{wcr}(T)$ of a given spanning tree $T \in \mathcal{T}$ is computed by a single minimum spanning tree computation under c^T , in $O(m \log n)$ time. In particular the inner maximisation of [\(4.1\)](#) is polynomial, and the decision version of [Section 4.1](#) lies in NP.*

Proof. By [Lemma 4.1](#), $\text{wcr}(T) = \sum_{e \in E(T)} u_e - \text{MST}(c^T)$. Assembling c^T costs $O(m)$ and the sum $O(n)$, while $\text{MST}(c^T)$ is one run of Kruskal’s algorithm, which computes a minimum spanning tree under any fixed cost vector in $O(m \log n)$ time ([Section 2.3](#)); this term dominates. Membership in NP then follows as in [Section 4.1](#), with T itself as the certificate. $\square$

What [Corollary 4.1](#) does not deliver is the second answer. [Corollary 3.1](#) settled both questions at once; here every tree can be evaluated, but none is singled out.

Worked Example. We apply [Lemma 4.1](#) to the micro-graph of [Figure 2.2](#), taking the trees one at a time, as the lemma prescribes a scenario for each.

For $T_1 = \{e_1, e_2, e_3\}$ the scenario sets e_1, e_2 and e_3 to their upper bounds and e_4, e_5 to their lower bounds, so $c^{T_1} = (4, 5, 7, 4, 5)$. Under it T_1 costs 16 and a minimum spanning tree costs 13, attained both by $\{e_1, e_2, e_4\}$ and by $\{e_1, e_4, e_5\}$, so $\text{wcr}(T_1) = 16 - 13 = 3$. The tie does no harm: the lemma fixes the scenario, not the tree the scenario is measured against.

For $T_2 = \{e_1, e_2, e_4\}$ the roles of e_3 and e_4 swap, giving $c^{T_2} = (4, 5, 1, 6, 5)$, and the same computation yields $\text{wcr}(T_2) = 15 - 10 = 5$, the minimum again attained twice. [Figure 4.1](#) sets the two scenarios side by side: e_3 swings furthest, from 7, charged to T_1 , which uses it, to 1, offered to any tree that can undercut T_2 ; both trees attaining the minimum of 10 take it.

[Table 4.1](#) adds T_3 and collects the three computations. Read together, the three scenarios obey one rule: any two of them differ exactly on the edges that lie in one of the two trees but not the other, and agree elsewhere, at an upper bound on a shared edge and at a lower bound on one both trees avoid.

Enumerating all eight spanning trees gives maximum regrets 3, 4, 5, 7, 8, 9, 9 and 10; the optimum is 3, attained by T_1 alone. How that optimum was found bears noting. In [Section 3.2](#) one minimum spanning tree computation under u produced the min-max optimum and certified it in the same stroke. Here [Corollary 4.1](#) evaluates any tree presented to it, but nothing points to the tree worth evaluating, and the optimum above was found by evaluating all eight. On eight trees that is an option; on a graph with exponentially many spanning trees it is not, and [Section 4.2.2](#) shows that no efficient substitute is to be expected.

Table 4.1: Lemma 4.1 on the three representative spanning trees of Figure 2.2. Each row solves one deterministic minimum spanning tree problem under its own scenario c^T ; the smallest maximum regret is set in bold.

T	c^T	$c^T(T)$	$\text{MST}(c^T)$	$\text{wcr}(T)$
T_1	$(4, 5, 7, 4, 5)$	16	13	3
T_2	$(4, 5, 1, 6, 5)$	15	10	5
T_3	$(2, 5, 7, 4, 7)$	19	11	8

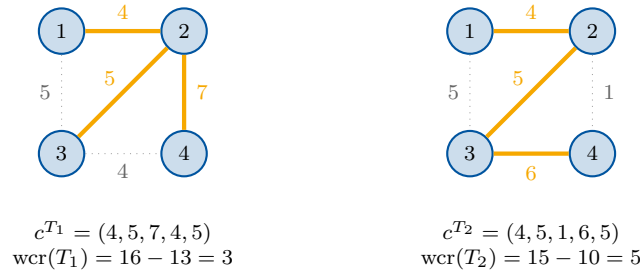


Figure 4.1: The scenarios of Lemma 4.1 for T_1 (left) and T_2 (right). Thick orange edges belong to the tree and carry their upper bounds; thin dotted edges lie outside it and carry their lower bounds. Between the panels only e_3 and e_4 change roles, and e_3 swings from 7 to 1.

The Role of the Lower Bounds. Lemma 3.1 never consulted the lower bounds; Lemma 4.1 consults both endpoints of every interval. Two points are worth drawing out.

First, the lower bounds enter through the second of the two rules alone, on the edges outside T , and that placement carries the chapter’s central contrast. Because the two rules split E into $E(T)$ and $E \setminus E(T)$, the scenario c^T moves as T moves: under the min-max objective the single scenario u served every tree, while under the regret objective each tree brings its own worst case.

Second, the tree-dependence is what closes the route of Corollary 3.1, which minimised over all trees at one fixed cost vector. The failure is not marginal. Evaluated at u , the regrets of T_1 and T_2 read 1 and 0 against the true maxima 3 and 5 of Table 4.1; indeed u understates the maximum regret of every spanning tree of the micro-graph. This is the first half of what the close of Section 3.2 promised: the regret comparison does depend on the lower bounds. The second half, that the interval problem becomes substantially harder, is the subject of Section 4.2.2.

4.2.2 Hardness and Approximation

The second question of Section 4.1 divides in two, and the halves pull against each other. Minimising the maximum regret over all spanning trees is strongly NP-hard, and yet a single minimum spanning tree computation at the right scenario comes within a factor of two of the optimum. Neither result is proved here; both are explained in enough detail that the mechanism, and not only the statement, is available.

The Source of the Hardness. A natural first hope is that the difficulty is numerical, and that instances with small or structured costs behave better. The opposite is true: the problem stays hard when all the intervals are identical.

Theorem 4.1 (Strong NP-Hardness under Interval Uncertainty). *Under interval uncertainty, the min-max regret spanning tree problem is strongly NP-hard, and it remains so when every cost interval equals $[0, 1]$.*

The result is due to Aron and Van Hentenryck [17] and Averbakh and Lebedev [18], in work published simultaneously [5, Theorem 8.16]. Its content becomes visible on the unit intervals. There the scenario of Lemma 4.1 charges each edge of T one and every other edge nothing, so any spanning tree S costs exactly the number of edges it shares with T , and

$$\text{wcr}(T) = (n - 1) - \min_{S \in \mathcal{T}} |E(T) \cap E(S)| = \max_{S \in \mathcal{T}} |E(T) \setminus E(S)|,$$

the largest number of its own edges that some spanning tree avoids. A spanning tree for which this maximum is as small as possible is called a *central tree* of the graph, a notion going back to Deo [19]. Finding one is itself strongly NP-hard [20]; see [5, Theorem 8.15]. The unit-interval case of the regret problem is therefore not merely reducible to the central tree problem; it is that problem. Both hardness proofs run through this object, one of them without naming it [5, Section 8.2.2].

The passage to $[0, 1]$ leaves no numerical magnitude to exploit, which is what makes the hardness strong in the sense of Section 2.5: no pseudo-polynomial algorithm exists unless $P = NP$. On the micro-graph the difficulty cannot be felt at all: five edges cannot hold two disjoint spanning trees, so every tree shares an edge with every other, and no maximum in the display above can exceed 2. All eight in fact equal 2, so every tree is central and nothing separates one candidate from another. The hardness has not closed the problem to exact computation: branch-and-bound algorithms have been developed for it [21].

The Midpoint Guarantee. With no efficient exact algorithm to be expected, the natural fallback is to trust one representative scenario and take a minimum spanning tree there. Section 4.2.1 has already disqualified the obvious candidate: at the all-upper-bounds vector the micro-graph’s maximum regrets read 1 for T_1 against 0 for T_2 , so that scenario would return T_2 , whose maximum regret is the larger of the two. The midpoints are the even-handed choice, and for them the fallback carries a guarantee.

Theorem 4.2 (2-Approximation via Midpoint). *Let $\mathcal{U} = \prod_{e \in E} [\ell_e, u_e]$ be an interval uncertainty set and let c^{av} denote the midpoint cost vector, with $c_e^{\text{av}} = \frac{1}{2}(\ell_e + u_e)$ for every edge $e \in E$. Every minimum spanning tree under c^{av} has maximum regret at most twice that of an optimal tree.*

The result is due to Kasperski and Zieliński [4], and [5, Theorem 5.26] states it for interval uncertainty over an arbitrary family of feasible solutions, spanning trees among them. The method is one line: assemble c^{av} and run Kruskal’s algorithm, at a cost of $O(m \log n)$.

Why the Midpoints. Every interval has a midpoint and a half-width, and the half-width is exactly the adversary's freedom: from the midpoint it may push a cost that far up, or that far down, and no further. In those terms the maximum regret splits in two. Set a tree T against a single rival S . The adversary's best answer raises the edges that only T uses and drops those that only S uses, and what it gains is the sum of two quantities. The first is the midpoint gap, the amount by which T is already dearer than S at the midpoints. The second is the room, the total half-width of the edges on which the two trees differ. The maximum regret of T is the largest value that gap and room together take, as S ranges over the spanning trees.

The choice of tree affects the two very differently. It can do little about the room: whichever tree is chosen, some rival differs from it on edges that have width, and that width is fixed by the intervals. The midpoint gap is another matter, because it is settled before the adversary moves at all. Choosing a minimum spanning tree under c^{av} makes it zero or negative against every rival, since no rival is cheaper at the midpoints. Such a tree therefore never starts behind, and nothing in its maximum regret comes of being dearer at the midpoints. What is left is the room, which no tree escapes. That this alone buys a factor of two is the content of [Theorem 4.2](#); [\[5, Lemmas 5.24 and 5.25\]](#) supply the two estimates behind it.

Worked Example. On the micro-graph the midpoint vector is $c^{\text{av}} = (3, 4, 4, 5, 6)$, which is the midpoint scenario $c^{(3)}$ of [Table 2.1](#), and the half-widths are $(1, 1, 3, 1, 1)$, so e_3 , the one wide interval, is where most of the room sits. Kruskal's algorithm selects T_1 under c^{av} , uniquely and at cost 11, and [Table 4.1](#) has already evaluated it: $\text{wcr}(T_1) = 3$, the optimum identified in [Section 4.2.1](#). The two pieces show in the numbers. Measured against T_2 , one of the two rivals at which its maximum is attained, T_1 carries a midpoint gap of -1 and room 4, and $-1 + 4$ is the 3 just recalled. The poorest of the eight, $\{e_2, e_4, e_5\}$, carries a gap of $+4$ and room 6, and its maximum regret is 10. Across the eight trees T_1 is the only one whose gap is never positive, which is precisely what choosing at the midpoints secures; the room it cannot touch, and what its maximum regret comes to is that room less the margin it holds at the midpoints. The guarantee promised no more than a factor of two; that the midpoint tree lands on the optimum exactly is a property of this instance, not of the method.

The interval case now closes in symmetry. Under the min-max objective, one minimum spanning tree computation at the upper bounds solved the problem outright ([Corollary 3.1](#)); under the regret objective, one at the midpoints comes within a factor of two, and [Theorem 4.1](#) is what rules the exact analogue out. Each objective ends at a single representative scenario; the difference is what one scenario can still deliver.

4.3 Discrete Scenarios

Under a finite uncertainty set the first of the two questions of [Section 4.1](#) needs no argument: the maximum in [\(4.2\)](#) runs over K scenarios and is read off directly, one difference per scenario. Only the second question remains, and the answer repeats the pattern of [Section 3.3](#). The problem is weakly NP-hard for every constant K , with a pseudo-polynomial algorithm and an approximation scheme alongside it, and

strongly NP-hard once K is part of the input. The repetition is no coincidence. Twice a construction of [Chapter 3](#) is reused outright, and the one reduction that is new grows out of a familiar grid.

4.3.1 Constant K

The natural first move is to reuse the two-scenario reduction of [Chapter 3](#), but in general a reduction built for one objective does not serve the other. The regret of a tree is its cost less the floor of its scenario. Subtracting a quantity that changes with the scenario can move the maximum to a different scenario, and with it the ranking of the trees. One case removes the difference entirely. If every scenario of an instance admits a spanning tree of cost zero, then every floor is zero and each difference $c^{(k)}(T) - \text{MST}(c^{(k)})$ collapses to $c^{(k)}(T)$. Maximum regret and worst-case cost then agree on every tree of that instance. The instances built for [Theorem 3.1](#) have exactly this property, so the reuse is sound there, and it yields the first result of the section.

Theorem 4.3 (Hardness for Two Scenarios). *Under discrete uncertainty, the min-max regret spanning tree problem is weakly NP-hard, even when restricted to $K = 2$ scenarios.*

The result is [[5](#), Theorem 8.7], obtained there from the construction of [Theorem 3.1](#), itself due to Kouvelis and Yu [[1](#)].

Proof of Theorem 4.3. Take the grid graph of [Figure 3.1](#) and its two scenarios without any alteration, so the construction stays computable in polynomial time. They encode a partition instance with weights $w_1, \dots, w_n$, total weight W and target $Q = W/2$.

Step 1: each scenario admits a spanning tree of cost zero. Step 2 of the proof of [Theorem 3.1](#) describes the spanning trees that contain every vertical edge: each is a set T_P , adding the first-row edge a_j for every $j \in P$ and the second-row edge b_j for every j outside P . Its two costs are $c^{(1)}(T_P) = w(P)$ and $c^{(2)}(T_P) = W - w(P)$. Choosing $P = \emptyset$ gives a spanning tree of cost zero in $c^{(1)}$, and $P = \{1, \dots, n\}$ one of cost zero in $c^{(2)}$. No cost in the construction is negative, so $\text{MST}(c^{(1)}) = \text{MST}(c^{(2)}) = 0$.

Step 2: the two objectives coincide on these instances. With both floors at zero, $\text{Regret}(T, c^{(k)}) = c^{(k)}(T)$ for every spanning tree T and each $k \in \{1, 2\}$, so $\text{wcr}(T) = \text{wc}(T)$ throughout. Step 3 of the same proof shows that the smallest worst-case cost equals Q when the partition instance is a yes-instance and exceeds Q otherwise, and the smallest maximum regret now inherits both statements. Deciding whether some spanning tree has maximum regret at most Q therefore decides the partition instance, and the problem is NP-hard. The hardness is weak rather than strong, since [Theorem 4.4](#) gives a pseudo-polynomial algorithm ([Section 2.5](#)). $\square$

A Small Instance. The instance worked in [Section 3.3](#) makes the coincidence concrete, its numbers intact. For $w = (3, 1, 2)$ the two trees of Step 1 are $T_\emptyset$ and $T_{\{1,2,3\}}$: each places the whole weight $W = 6$ in one scenario and nothing in the other, so both floors are zero. The costs $c^{(1)}(T_{\{1\}}) = c^{(2)}(T_{\{1\}}) = 3$ recorded there are therefore also its two regrets, and the resulting maximum regret of $3 = Q$ is the best any spanning tree achieves. For $w = (1, 5)$ the worst-case costs 5 and 6 become

maximum regrets in the same way. The smaller of them lies above $Q = 3$, matching the fact that no subset of $\{1, 5\}$ sums to 3.

The positive half of the constant- K picture crosses over just as directly.

Theorem 4.4 (Pseudo-Polynomial Algorithm and FPTAS). *For every constant K , the min-max regret spanning tree problem under discrete uncertainty is solvable in pseudo-polynomial time and admits a fully polynomial-time approximation scheme.*

Both results are due to Aissi, Bazgan and Vanderpooten [12] and are stated together as [5, Theorem 8.6]. The route is the one taken in Section 3.3, with a single adjustment. There a tree carried a list of K costs, the search ran over the lists that some tree realises, and the exact spanning tree problem tested each prescribed list. Regret shifts every entry of such a list by the floor of its scenario, and the floors do not depend on the tree: the K minimum spanning trees are computed once, as Section 4.1 has already noted. Subtracting K fixed numbers leaves the realisable lists a translate of the old collection, pseudo-polynomially many as before. The same search therefore returns the tree whose largest entry is smallest, the entries now being regrets, and the same scaling delivers the approximation scheme. Both guarantees carry the same hypothesis as their min-max counterparts, that K is fixed. Section 4.3.2 takes that hypothesis away.

4.3.2 Unbounded K

With K part of the input the hardness becomes strong here as well, but it is not reached the way Section 3.3 reached it. There the route to strong hardness left the partition problem behind and started again from 3-SAT. Here it stays with weights and strengthens the problem they come from.

Theorem 4.5 (Strong NP-Hardness and Inapproximability). *If K is part of the input, the min-max regret spanning tree problem under discrete uncertainty is strongly NP-hard, and unless $P = NP$ it cannot be approximated within a factor of $2 - \varepsilon$ for any $\varepsilon > 0$.*

The hardness is due to Kouvelis and Yu [1] and the inapproximability to Kasperski and Zieliński [13]; the two appear as [5, Theorems 8.8 and 8.9].

The hardness rests on the *3-partition problem*, a harder relative of the partition problem behind Theorem 3.1. An instance gives $3m$ positive integers with total mB , each of them strictly between $B/4$ and $B/2$, and asks whether they split into m triples of sum B . Its numbers can be kept small: the problem stays NP-hard even when every one of them is bounded by a polynomial in the input [22, Section 4.2.2], which is strong hardness in the sense of Section 2.5. The scaling that tamed the constant- K case therefore has nothing left to shrink. The reduction writes a 3-partition instance into the grid of Figure 3.1, grown from two rows to m , with one scenario charging each row. A spanning tree of cost zero is again available in every scenario, so the two objectives coincide here just as they did in Section 4.3.1.

The bound of $2 - \varepsilon$ needs no new construction at all. It is inherited from Theorem 3.4, and the zero floor does its work a third time. Every scenario of that construction charges two literal edges and leaves all other edges free, while each gadget offers three literal routes. No scenario can therefore block every route of a

gadget. The tree built in that proof takes one literal edge per gadget, and for a given scenario those edges can be chosen among the routes it leaves free, so the tree costs nothing at all. The floor is zero in every scenario once more, maximum regret equals worst-case cost, and the argument that reads satisfiability off the returned tree carries over unchanged. The problem admits no constant-factor approximation at all [14], exactly as under the min-max objective.

The three hardness results therefore rest on a single observation, the zero floors of [Section 4.3.1](#): each of their constructions admits a spanning tree of cost zero in every scenario, so regret there can be measured as cost. The positive side is where the mirror breaks. No regret counterpart of the logarithmic guarantee of [Theorem 3.5](#) is known. Whether the argument behind that guarantee can be adapted is on record as an open question [5, Open Problem 1].

Summary

The two questions of [Section 4.1](#) organised this chapter: what form the maximum regret takes as a function of the tree, and whether that form can be minimised over all trees at once. The first is answered in both models, at very different cost. Under discrete uncertainty it costs nothing, the maximum running over finitely many scenarios. Under interval uncertainty it takes [Lemma 4.1](#): the maximum is attained at a scenario read off the tree itself, so each tree brings its own worst case.

A given tree can be evaluated in polynomial time under either model; all of the difficulty lies in choosing one. Over an interval set the choice is strongly **NP**-hard, and stays so when every interval is identical: no magnitude is left for an algorithm to work on. Over discrete scenarios it is weakly **NP**-hard already at two, and turns strong only once their number is free to grow.

[Chapter 3](#) closed on a criterion: whether the adversary’s dependence on the tree can be stripped away by a bounded amount of extra work. Two of its three models met it, intervals needing no work at all, since one cost vector was worst for every tree, and a budget needing one enumerated parameter. Neither model of this chapter meets it unless $\mathbf{P} = \mathbf{NP}$, and under discrete scenarios neither does the min-max objective. Under intervals it did, and losing that is the price of judging a tree against the best one available in its scenario rather than by an absolute standard.

The interval extremal characterisation and the two-scenario reduction are proved in full here. The interval hardness, the midpoint guarantee and the transfer of the inapproximability bound are developed from their sources in the text. Only the algorithms for constant K and the strong hardness for unbounded K are stated on the authority of the literature. [Chapter 5](#) sets the two objectives side by side across all their models, including the one left aside here.

Chapter 5

Conclusion

Chapters 3 and 4 each followed one objective across the uncertainty models of Section 2.4. This chapter sets the two side by side. Section 5.1 condenses both chapters into one classification and states what it shows. Section 5.2 solves the running micro-graph under the objectives and models the thesis develops. Section 5.3 turns to the models and questions beyond the scope of this thesis. Every claim rests on a result established earlier or on a source cited at the point of use; nothing further is proved.

5.1 Synthesis of Results

Complexity Landscape. Table 5.1 holds the classification. Each column is one uncertainty model, with the discrete model split into constant K and K part of the input, because the two sides are classified differently. Each objective fills two rows: the complexity of the problem, and the algorithms available for it. Every entry names the result or source that establishes it.

Five of the six combinations of objective and model are developed in Chapters 3 and 4. The sixth, min-max regret under a budget, was left aside by Chapter 4 and is settled here by reduction from the interval case; the note beneath the table gives the argument.

Key Patterns. The discrete column does not separate the two objectives. Min-max and min-max regret carry the same complexity at every K , and while K is constant they carry the same algorithms as well. Both are weakly NP-hard already at two scenarios, solvable in pseudo-polynomial time and approximable to any precision. Both turn strongly NP-hard once K is part of the input, and unless $P = NP$ neither is approximable within $2 - \varepsilon$. Beyond two scenarios, K decides only whether that hardness is weak or strong. The interval and budgeted columns separate the objectives completely: under each of them min-max is solved exactly in polynomial time, while min-max regret is strongly NP-hard. These two separations are not independent. The regret hardness that creates the budgeted one is inherited from the interval one, by the reduction in the note beneath the table.

At constant K both objectives admit an FPTAS, the strongest guarantee short of exactness. Every remaining guarantee in the table but one comes from a single method: solve the deterministic problem at an averaged cost vector. The midpoint of

Table 5.1: Classification of the robust spanning tree problem, by objective and by uncertainty model. The entry “no $2 - \varepsilon$ ” means that no polynomial-time approximation within a factor of $2 - \varepsilon$ exists for any $\varepsilon > 0$, unless $P = NP$. The note beneath the table explains the two entries marked † .

	Discrete scenarios			
	Constant K	K in the input	Interval	Budgeted
<i>Min-max</i>				
Complexity	weakly NP-hard (Theorem 3.1)	strongly NP-hard; no $2 - \varepsilon$ (Theorem 3.4)	in P (Corollary 3.1)	in P (Theorem 3.6)
Algorithms	pseudo-polynomial; FPTAS (Theorems 3.2 and 3.3)	$O(\log K)$ - approximation (Theorem 3.5)	exact, $O(m \log n)$ (Corollary 3.1)	exact, $O(m^2 \log n)$ (Theorem 3.6)
<i>Min-max regret</i>				
Complexity	weakly NP-hard (Theorem 4.3)	strongly NP-hard; no $2 - \varepsilon$ (Theorem 4.5)	strongly NP-hard (Theorem 4.1)	strongly NP-hard †
Algorithms	pseudo-polynomial; FPTAS (Theorem 4.4)	K -approximation [5, Theorem 5.4]	2-approximation (Theorem 4.2)	none developed †

† The hardness entry follows by reduction from the interval column. At $\Gamma = |E|$ the budget constraint is inactive, and the budgeted set of Definition 2.3 is exactly the interval set $\prod_{e \in E} [\bar{c}_e, \bar{c}_e + \hat{c}_e]$. The choice $\bar{c} = \ell$, $\hat{c} = u - \ell$ therefore writes any interval regret instance as a budgeted one. Goerigk and Hartisch [5, Section 8.5] record the same inheritance. The hardness inherited is strong, because Theorem 4.1 holds already when every interval equals $[0, 1]$, where every number of the instance is a 0 or a 1. The algorithm entry stays empty because this thesis develops none. Under interval uncertainty the maximum regret of a single tree is found by one minimum spanning tree computation (Corollary 4.1). Under a budget no such shortcut is known in general, and the reformulation of Goerigk and Hartisch [23] still optimises over all spanning trees [5, Theorem 4.25].

each interval gives the 2-approximation for interval regret (Theorem 4.2); the average of the K scenarios gives the K -approximation for both objectives [5, Theorem 5.4]. The exception is the $O(\log K)$ bound for min-max (Theorem 3.5), for which no regret counterpart is known (Section 4.3.2); this asymmetry, and the question whether the factor 2 is the best possible, are taken up in Section 5.3.

What decides this classification is not the size of the uncertainty set: an interval set is infinite yet polynomial for min-max, while two scenarios already make both objectives hard. The deciding factor, identified in the Summary of Chapter 3 and extended in that of Chapter 4, is whether the adversary’s worst case depends on the chosen tree, and at what cost that dependence can be removed. Min-max removes it under both continuous models, at no cost under intervals and at the cost of enumerating one level under a budget; these are the table’s two polynomial cells (Corollary 3.1 and Theorem 3.6). For regret the worst case moves with the tree instead, a dependence Section 4.2.1 makes explicit for intervals. Wherever the table records hardness, on either objective, no such bounded removal can exist unless $P = NP$. The answer to the question this thesis opened with is therefore short: across objectives and models, complexity is set by the cost of making the adversary’s answer independent of the tree.

5.2 Example Gallery

Micro-Graph Solutions Across Objectives. Table 5.1 says how hard each problem is; it does not show what the objectives do. This section shows them at work, on the instance that has run through the thesis since Figure 2.2. The upper panel of Figure 5.1 restates its data, five edges with interval costs and the three discrete scenarios read off them, so that the values below can be traced back to it. The grid then solves each combination the thesis develops and reports the optimal tree, which is unique in every case, together with its optimal value.

The two rows do not mix. Min-max returns T_2 in each of its three panels, and min-max regret returns T_1 in each of its two. On this instance the choice of objective decides the answer, while the choice of uncertainty model changes only the value attached to it. One of these agreements is built into the instance rather than found by the objectives. $c^{(2)}$ is the vector of upper bounds, and it is at least as expensive as the other two scenarios on every tree. Panels (a) and (b) therefore pose the same problem, and they agree on all eight trees, not only at the optimum.

Figure 5.2 explains why the contest is so narrow, and what choosing between its two winners amounts to. Six of the eight trees land in the shaded region, worse than T_1 in both of the quantities this thesis has minimised, so neither objective can ever return them. What remains is a plain trade: T_2 saves one unit of worst-case cost, T_1 saves two units of maximum regret, and no tree offers both. Drawn with the discrete values instead, the picture keeps the same two trees outside the region and the same trade of one against two.

One more answer belongs beside these two: the tree a planner would build with no robustness at all. Reading every interval at its midpoint and solving the deterministic problem returns T_1 , at cost 11, so it coincides with the regret answer and not with the min-max one. The comparison has content only because the midpoint reads the intervals evenly: read at their upper bounds instead, the deterministic problem is

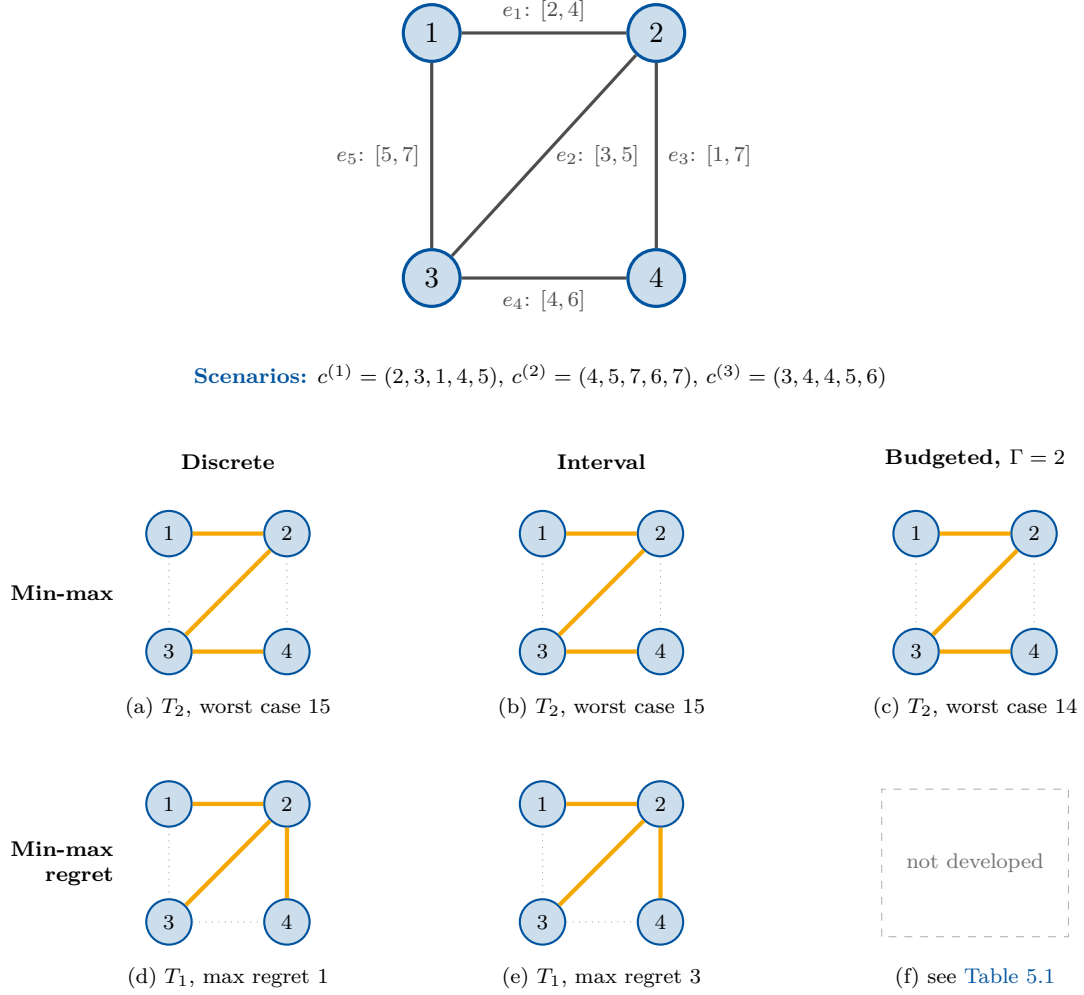


Figure 5.1: The running micro-graph solved under each objective and each uncertainty model. The upper panel restates the instance: the interval costs, and the three discrete scenarios derived from them as their lower bounds, upper bounds and midpoints. In the grid, orange edges form the optimal tree, which is unique in every panel; the remaining edges are dotted. The upper row reports worst-case cost and the lower row maximum regret, so values are comparable within a row only. Panel (f) carries no tree by design: no algorithm for it is developed in this thesis, and its hardness is inherited from the interval case through the note to [Table 5.1](#).

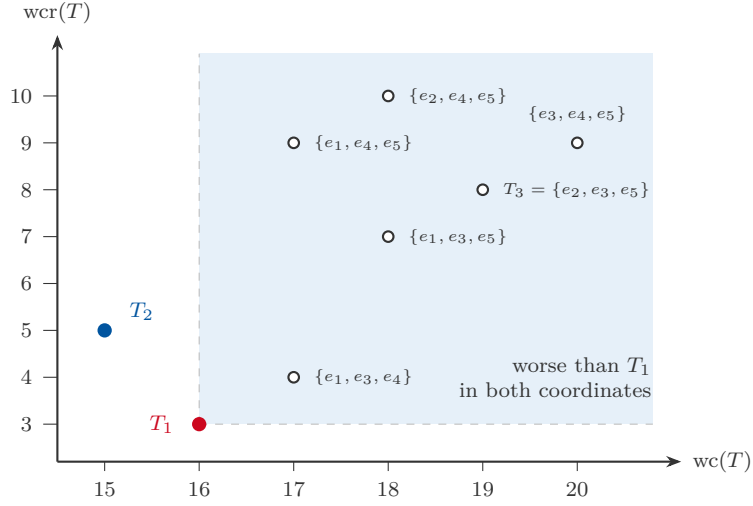


Figure 5.2: Why the contest in Figure 5.1 is between two trees only. Each of the eight spanning trees is placed by its worst-case cost $wc(T)$ and its maximum regret $wcr(T)$ under interval uncertainty. The shaded region is anchored at T_1 and holds every tree that is worse than T_1 in both coordinates.

the min-max interval problem itself (Corollary 3.1). Under interval costs min-max is therefore the deterministic answer to the most pessimistic reading of the data, whatever the instance. That regret agrees with the deterministic answer to the even reading is a fact about this instance alone.

5.3 Scope and Outlook

Extensions in Uncertainty Modelling. The nearest untreated problem sits inside Table 5.1 itself: min-max regret under a budget. It was left aside not because nothing is known about it, but because almost nothing is. Its hardness follows in one step from the interval case, and there the record ends; the overview of Goerigk and Hartisch [5, Table 8.1] lists no algorithm and no guarantee for the problem. This thesis set out to curate and prove what is established, and for this combination the established part is exactly the note beneath its table. The budgeted model invites a second question of its own. A single number Γ caps the cumulative deviation across the whole graph, wherever the adversary chooses to place it; a network whose regions face risks of their own invites one cap per region instead. Theorem 3.6 does not extend as it stands. It trades the budget for a single level, which takes one of at most $m + 1$ values, and pays one minimum spanning tree computation at each of them. With several budgets there is a level to fix for each, and the candidates multiply. Whether some other device restores the polynomial result is not settled by anything in this thesis.

Extensions in Solution Concepts. Every problem in Table 5.1 rests on one assumption that no choice of cost model touches: the whole tree is fixed before a single cost is known. Two settings in the literature relax it. In the two-stage problem, part of the tree is bought at known first-stage costs and the rest is added once the scenario is revealed, at the costs it brings [5, Definition 3.3]. In the recoverable

problem, a complete tree is chosen at once, with the right to exchange a bounded number of its edges after the scenario is known [5, Definition 3.4].

Under discrete scenarios, flexibility does not buy tractability. The two-stage problem is NP-hard for constant K , already at two scenarios, as Goerigk, Kasperski and Zieliński [24] show [5, Theorem 8.17]. Kasperski, Kurpisz and Zieliński [25] prove the same for the recoverable problem, whatever the number of exchanges the recovery allows [5, Theorem 8.19]. In both, the hardness turns strong once K is part of the input [5, Theorems 8.18 and 8.20]. That is the same line Table 5.1 draws in its own discrete column.

Under interval costs, flexibility keeps it. The two-stage problem has the same complexity as its deterministic counterpart [5, Theorem 4.10]. The recoverable problem is solvable in polynomial time as well [5, Corollary 8.22], a result Hradovich, Kasperski and Zieliński [26] establish for spanning trees directly. The split that organised Chapter 3, hard under discrete scenarios and tractable under interval costs, is therefore not a property of the min-max objective alone: it persists when the tree itself is allowed to move.

Open Problems. Two questions from the literature run through this thesis, and both remain open exactly where its table stops. The first asks whether any algorithm can improve on the factor 2 for a min-max regret problem with interval costs whose deterministic version is polynomial [5, Open Problem 2]. An answer would settle the one cell of Table 5.1 in which a constant factor is known but its optimality is not. The second asks whether the technique behind the logarithmic min-max guarantee extends to the regret objective [5, Open Problem 1]. An answer would close the one gap in Table 5.1 where the two objectives share a complexity but not a guarantee.

What this thesis leaves behind is deliberately small and thoroughly checked: a classification whose every entry is carried by a proof or by a cited source, a criterion that reads it, and a five-edge graph on which the disagreement of the two objectives can be verified by hand. A planner who must commit to a network before its costs are certain still faces the choice this thesis began with, between guarding against the worst outcome and guarding against regret. What such a planner no longer has to guess is what either choice will cost to compute.

Appendix A

Notation

This appendix provides a reference for the mathematical notation used throughout the thesis. Symbols are grouped by category, with the section in which each symbol is introduced.

Symbol	Meaning	Introduced
<i>Graph Notation</i>		
$G = (V, E)$	Undirected graph with vertex set V , edge set E	§2.1
$n = V , m = E $	Number of vertices and edges	§2.1
$\delta(X)$	Cut induced by vertex set $X \subseteq V$	§2.1
<i>Trees and Costs</i>		
$\mathcal{T}$	Set of all spanning trees of G	§2.1
T	Spanning tree, $T \in \mathcal{T}$	§2.1
$E(T)$	Edge set of the spanning tree T , with $ E(T) = n - 1$	§2.1
$c: E \rightarrow \mathbb{R}$	Edge cost function	§2.1
$c(T)$	Cost of tree T , the sum $\sum_{e \in E(T)} c_e$	§2.1
$\text{MST}(c)$	Minimum spanning tree cost under c	§2.1
T^*	A minimum spanning tree, $c(T^*) = \text{MST}(c)$	§2.1
C_f	Fundamental cycle of non-tree edge f	§2.1
$\delta(X_e)$	Fundamental cut of tree edge e	§2.1
<i>Uncertainty Models</i>		
$\mathcal{U}$	Uncertainty set (discrete, interval, or budgeted)	§2.4
K	Number of discrete scenarios	§2.4
$c^{(k)}$	Scenario k cost vector, $k \in \{1, \dots, K\}$	§2.4
$[\ell_e, u_e]$	Interval cost bounds for edge e	§2.4
ℓ_e, u_e	Lower and upper bounds on edge cost (interval model)	§2.4
$\mathcal{U}^\Gamma$	Budgeted uncertainty set	§2.4

Continued on next page

Symbol	Meaning	Introduced
Γ	Deviation budget (budgeted model)	§2.4
$\bar{c}_e$	Nominal cost of edge e (budgeted model)	§2.4
$\hat{c}_e$	Maximum deviation of edge e (budgeted model)	§2.4
δ_e	Fraction of its maximum deviation taken by edge e	§2.4
$\mathbb{R}_{\geq 0}^{ E }$	Non-negative cost vectors, required by all three models	§2.4
$u = (u_e)_{e \in E}$	Upper-bound cost vector, to which the interval model reduces	§3.2
c^T	Scenario charging u_e on $E(T)$ and ℓ_e elsewhere, at which the maximum regret of T is attained	§4.2.1
c^{av}	Midpoint cost vector, $c_e^{\text{av}} = \frac{1}{2}(\ell_e + u_e)$	§4.2.2
π	Level above which deviations are charged (budgeted reformulation)	§3.4
Π	Candidate levels $\{0\} \cup \{\hat{c}_e : e \in E\}$	§3.4
$[z]^+$	Positive part, $\max\{z, 0\}$	§3.4
<i>Robust Objectives</i>		
$\text{Regret}(T, c)$	Regret of T under c : $c(T) - \text{MST}(c)$	§2.4
$\text{wc}(T)$	Worst-case cost of tree T : $\max_{c \in \mathcal{U}} c(T)$	§3.1
$\text{wcr}(T)$	Maximum regret of tree T : $\max_{c \in \mathcal{U}} \text{Regret}(T, c)$	§4.1
<i>Complexity</i>		
P	Polynomial-time complexity class	§2.5
NP	Nondeterministic polynomial-time class	§2.5
NP-hard	At least as hard as NP-complete problems	§2.5
Weakly NP-hard	NP-hard with a pseudo-polynomial algorithm	§2.5
Strongly NP-hard	NP-hard even with poly-bounded numeric values	§2.5
Pseudo-polynomial	Algorithm polynomial in input size and numeric magnitudes	§2.5
α -approximation	Algorithm guaranteeing $\text{ALG} \leq \alpha \cdot \text{OPT}$	§2.5
ε	Tolerance parameter for approximation schemes	§2.5
PTAS	Polynomial-time approximation scheme	§2.5
FPTAS	Fully polynomial-time approximation scheme	§2.5

Bibliography

- [1] Panos Kouvelis and Gang Yu. *Robust Discrete Optimization and Its Applications*. Vol. 14. Nonconvex Optimization and Its Applications. Dordrecht: Kluwer Academic Publishers, 1997. ISBN: 978-0-7923-4291-5. DOI: [10.1007/978-1-4757-2620-6](https://doi.org/10.1007/978-1-4757-2620-6).
- [2] Bernhard Korte and Jens Vygen. *Combinatorial Optimization: Theory and Algorithms*. 6th ed. Vol. 21. Algorithms and Combinatorics. Berlin, Heidelberg: Springer, 2018. ISBN: 978-3-662-56039-6. DOI: [10.1007/978-3-662-56039-6](https://doi.org/10.1007/978-3-662-56039-6).
- [3] Dimitris Bertsimas and Melvyn Sim. ‘Robust discrete optimization and network flows’. In: *Mathematical Programming* 98.1–3 (2003), pp. 49–71. DOI: [10.1007/s10107-003-0396-4](https://doi.org/10.1007/s10107-003-0396-4).
- [4] Adam Kasperski and Paweł Zieliński. ‘An approximation algorithm for interval data minmax regret combinatorial optimization problems’. In: *Information Processing Letters* 97.5 (2006), pp. 177–180. DOI: [10.1016/j.ipl.2005.11.001](https://doi.org/10.1016/j.ipl.2005.11.001).
- [5] Marc Goerigk and Michael Hartisch. *An Introduction to Robust Combinatorial Optimization: Concepts, Models and Algorithms for Decision Making under Uncertainty*. Vol. 361. International Series in Operations Research & Management Science. Cham: Springer, 2024. DOI: [10.1007/978-3-031-61261-9](https://doi.org/10.1007/978-3-031-61261-9).
- [6] Joseph B. Kruskal. ‘On the shortest spanning subtree of a graph and the traveling salesman problem’. In: *Proceedings of the American Mathematical Society* 7.1 (1956), pp. 48–50. DOI: [10.1090/S0002-9939-1956-0078686-7](https://doi.org/10.1090/S0002-9939-1956-0078686-7).
- [7] Robert C. Prim. ‘Shortest connection networks and some generalizations’. In: *Bell System Technical Journal* 36.6 (1957), pp. 1389–1401. DOI: [10.1002/j.1538-7305.1957.tb01515.x](https://doi.org/10.1002/j.1538-7305.1957.tb01515.x).
- [8] Christina Büsing. ‘Combinatorial Optimization – Optimierung B (Skript OptiB)’. Lecture notes, Version: February 4, 2023. RWTH Aachen, 2023.
- [9] Ravindra K. Ahuja, Thomas L. Magnanti and James B. Orlin. *Network Flows: Theory, Algorithms, and Applications*. Upper Saddle River, NJ: Prentice Hall, 1993. ISBN: 978-0-13-617549-0.
- [10] Hassene Aissi, Cristina Bazgan and Daniel Vanderpooten. ‘Min–max and min–max regret versions of combinatorial optimization problems: A survey’. In: *European Journal of Operational Research* 197.2 (2009), pp. 427–438. DOI: [10.1016/j.ejor.2008.09.012](https://doi.org/10.1016/j.ejor.2008.09.012).

- [11] Francisco Barahona and William R. Pulleyblank. ‘Exact arborescences, matchings and cycles’. In: *Discrete Applied Mathematics* 16.2 (1987), pp. 91–99. DOI: [10.1016/0166-218X\(87\)90067-9](https://doi.org/10.1016/0166-218X(87)90067-9).
- [12] Hassene Aissi, Cristina Bazgan and Daniel Vanderpooten. ‘Approximation complexity of min-max (regret) versions of shortest path, spanning tree, and knapsack’. In: *Algorithms – ESA 2005: 13th Annual European Symposium*. Vol. 3669. Lecture Notes in Computer Science. Berlin, Heidelberg: Springer, 2005, pp. 862–873. DOI: [10.1007/11561071_76](https://doi.org/10.1007/11561071_76).
- [13] Adam Kasperski and Paweł Zieliński. ‘On the approximability of minmax (regret) network optimization problems’. In: *Information Processing Letters* 109.5 (2009), pp. 262–266. DOI: [10.1016/j.ipl.2008.10.008](https://doi.org/10.1016/j.ipl.2008.10.008).
- [14] Adam Kasperski and Paweł Zieliński. ‘On the approximability of robust spanning tree problems’. In: *Theoretical Computer Science* 412.4–5 (2011), pp. 365–374. DOI: [10.1016/j.tcs.2010.10.006](https://doi.org/10.1016/j.tcs.2010.10.006).
- [15] Werner Baak, Marc Goerigk, Adam Kasperski and Paweł Zieliński. ‘Robust min-max (regret) optimization using ordered weighted averaging’. In: *European Journal of Operational Research* 322.1 (2025), pp. 171–181. DOI: [10.1016/j.ejor.2024.10.028](https://doi.org/10.1016/j.ejor.2024.10.028).
- [16] Vittorio Bilò, Ioannis Caragiannis, Angelo Fanelli, Michele Flammini and Gianpiero Monaco. ‘Simple Greedy Algorithms for Fundamental Multidimensional Graph Problems’. In: *44th International Colloquium on Automata, Languages, and Programming (ICALP 2017)*. Vol. 80. Leibniz International Proceedings in Informatics (LIPIcs). Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2017, 125:1–125:13. DOI: [10.4230/LIPIcs.ICALP.2017.125](https://doi.org/10.4230/LIPIcs.ICALP.2017.125).
- [17] Ionuț D. Aron and Pascal Van Hentenryck. ‘On the complexity of the robust spanning tree problem with interval data’. In: *Operations Research Letters* 32.1 (2004), pp. 36–40. DOI: [10.1016/S0167-6377\(03\)00058-0](https://doi.org/10.1016/S0167-6377(03)00058-0).
- [18] Igor Averbakh and Vasilij Lebedev. ‘Interval data minmax regret network optimization problems’. In: *Discrete Applied Mathematics* 138.3 (2004), pp. 289–301. DOI: [10.1016/S0166-218X\(03\)00462-1](https://doi.org/10.1016/S0166-218X(03)00462-1).
- [19] Narsingh Deo. ‘A central tree’. In: *IEEE Transactions on Circuit Theory* 13.4 (1966), pp. 439–440. DOI: [10.1109/TCT.1966.1082617](https://doi.org/10.1109/TCT.1966.1082617).
- [20] Sergei Bezrukov, Firoz Kaderali and Werner Poguntke. ‘On central spanning trees of a graph’. In: *Combinatorics and Computer Science*. Vol. 1120. Lecture Notes in Computer Science. Berlin, Heidelberg: Springer, 1996, pp. 53–57. DOI: [10.1007/3-540-61576-8_73](https://doi.org/10.1007/3-540-61576-8_73).
- [21] Roberto Montemanni and Luca Maria Gambardella. ‘A branch and bound algorithm for the robust spanning tree problem with interval data’. In: *European Journal of Operational Research* 161.3 (2005), pp. 771–779. DOI: [10.1016/j.ejor.2003.10.008](https://doi.org/10.1016/j.ejor.2003.10.008).

- [22] Michael R. Garey and David S. Johnson. *Computers and Intractability: A Guide to the Theory of NP-Completeness*. A Series of Books in the Mathematical Sciences. San Francisco: W. H. Freeman and Company, 1979. ISBN: 0-7167-1045-5.
- [23] Marc Goerigk and Michael Hartisch. ‘Combinatorial optimization problems with balanced regret’. In: *Discrete Applied Mathematics* 328 (2023), pp. 40–59. DOI: [10.1016/j.dam.2022.11.013](https://doi.org/10.1016/j.dam.2022.11.013).
- [24] Marc Goerigk, Adam Kasperski and Paweł Zieliński. ‘Two-stage combinatorial optimization problems under risk’. In: *Theoretical Computer Science* 804 (2020), pp. 29–45. DOI: [10.1016/j.tcs.2019.10.035](https://doi.org/10.1016/j.tcs.2019.10.035).
- [25] Adam Kasperski, Adam Kurpisz and Paweł Zieliński. ‘Recoverable robust combinatorial optimization problems’. In: *Operations Research Proceedings 2012*. Cham: Springer, 2014, pp. 147–153. DOI: [10.1007/978-3-319-00795-3_22](https://doi.org/10.1007/978-3-319-00795-3_22).
- [26] Mikita Hradovich, Adam Kasperski and Paweł Zieliński. ‘The recoverable robust spanning tree problem with interval costs is polynomially solvable’. In: *Optimization Letters* 11.1 (2017), pp. 17–30. DOI: [10.1007/s11590-016-1057-x](https://doi.org/10.1007/s11590-016-1057-x).

Declaration on the Use of AI Tools

During the writing of this thesis I used Claude (Anthropic) in order to improve the readability and language of the manuscript. After using this tool/service, I reviewed and edited the content as needed and take full responsibility for the content of this thesis.